

Ultradifferentiable functions and Fourier analysis

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Classes of non-quasianalytic functions are classically defined by imposing growth conditions on the derivatives of the functions. It was Beurling [1] (see Björck [2]) who pointed out that decay properties of the Fourier transform of a compactly supported function can be used for this purpose equally well.

In the present article we modify Beurling's approach. More precisely, we call $\omega: [0, \infty[\rightarrow [0, \infty[$ a weight function if ω is continuous and satisfies

$$(\alpha) \quad \omega(2t) = O(\omega(t))$$

$$(\gamma) \quad \log t = o(\omega(t))$$

$$(\beta) \quad \int_1^\infty \frac{\omega(t)}{t^2} dt < \infty$$

$$(\delta) \quad \varphi: t \mapsto \omega(e^t) \text{ is convex.}$$

We show that for each open subset $\Omega \neq \emptyset$ of $\mathbb{R}^N$ the spaces

$$\mathcal{D}_{\{\omega\}}(\Omega) := \left\{ f \in C_c(\Omega) \mid \text{there is } \varepsilon > 0: \int_{\mathbb{R}^N} |\hat{f}(t)| e^{\varepsilon \omega(t)} dt < \infty \right\}$$

and

$$\mathcal{D}_{(\omega)}(\Omega) := \left\{ f \in C_c(\Omega) \mid \text{for all } k > 0: \int_{\mathbb{R}^N} |\hat{f}(t)| e^{k \omega(t)} dt < \infty \right\}$$

are nontrivial even if ω is only increasing and satisfies (α) – (γ) . The modification compared to Beurling-Björck is that the subadditivity of ω is replaced by the weaker condition (α) . Condition (δ) allows us to describe $\mathcal{D}_{\{\omega\}}(\Omega)$ and $\mathcal{D}_{(\omega)}(\Omega)$ in the following way

$$\mathcal{D}_{\{\omega\}}(\Omega) = \left\{ f \in \mathcal{D}(\Omega) \mid \text{there is } k \in \mathbb{N}: \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in \Omega} |f^{(\alpha)}(x)| \exp\left(-\frac{1}{k} \varphi^*(|\alpha|k)\right) < \infty \right\},$$

$$\mathcal{D}_{(\omega)}(\Omega) = \left\{ f \in \mathcal{D}(\Omega) \mid \text{for all } k \in \mathbb{N}: \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in \Omega} |f^{(\alpha)}(x)| \exp\left(-k \varphi^*\left(\frac{|\alpha|}{k}\right)\right) < \infty \right\},$$

where φ^* denotes the Young conjugate of the convex function $\varphi: t \mapsto \omega(e^t)$. Knowing this, the corresponding local spaces $\mathcal{E}_{\{\omega\}}(\Omega)$ and $\mathcal{E}_{(\omega)}(\Omega)$ can be introduced. Then routine arguments show that these classes—endowed with the usual locally convex topologies—establish a theory of ultradistributions in the sense of Ciorănescu and Zsidó [9], section 7. In fact, it is equivalent to the theory developed in [9] and strictly larger than the theory of Beurling-Björck. This follows from results of [9] and the observation of U. Franken that all the spaces of test functions used in [9] appear in the theory presented here.

Condition (δ) allows us to use Hörmander's L^2 -techniques when Fourier-Laplace transforms of ultradistributions with compact support are investigated. Therefore the theory

presented here provides a suitable and smooth frame whenever Fourier analysis is applied. In fact, it was used already successfully to clarify when the analogue of Whitney's extension theorem holds and when the extension can be given by a continuous linear operator (see Bonet, Braun, Meise, and Taylor [3], Bonet, Meise, and Taylor [4], Meise and Taylor [21], [22]). Moreover, it was applied in Meise [19] and in Braun, Meise, and Vogt [6], [7], to study convolution operators on $\mathcal{E}_{\{\omega\}}(\mathbb{R})$ and partial differential operators with constant coefficients on $\mathcal{E}_{\{\omega\}}(\mathbb{R}^N)$.

The article has eight sections. In the first two we prove—as basic ingredients—some technical lemmas on weight functions and on the existence of certain smooth functions with compact support. After that, well known arguments (which are mostly not reproduced) show that the classical results can also be derived in the present framework. More precisely, the spaces of test functions $\mathcal{D}_{\{\omega\}}(\Omega)$ and $\mathcal{D}_{\{\omega\}}(\Omega)$ are introduced in section three and the analogue of the Paley-Wiener theorem is proved. In section four the locally convex algebras $\mathcal{E}_{\{\omega\}}(\Omega)$ and $\mathcal{E}_{\{\omega\}}(\Omega)$ are introduced and a result of Vogt [27] is used to represent them as sequence spaces. The same is done for $\mathcal{D}_{\{\omega\}}(\Omega)'$ and $\mathcal{D}_{\{\omega\}}(\Omega)'$ in section five, while convolutions of test functions and ultradistributions are introduced in section six. In section seven, the analogue of the theorem of Paley-Wiener-Schwartz is proved and it is shown that for each weight function ω

$$\mathcal{D}_{\{\omega\}}(\Omega)' = \bigcup \left\{ \mathcal{D}_{\{\sigma\}}(\Omega)' \mid \sigma \text{ is a weight function with } \sigma = o(\omega) \right\}.$$

In the last section we indicate how $\mathcal{D}_*(K)$ and $\mathcal{E}_*(\Omega)$ behave under tensor products and we compare the present theories of ultradistributions with other existing theories.

The authors thank U. Franken for the communication of Lemma 8.5.

1 Weight functions

In this section we introduce the weight functions we will use in the subsequent sections. Moreover, we state and prove some technical results about existence of weight functions with certain upper and lower estimates.

1.1 Definition. Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be a continuous even function with $\omega|_{[0, 1]} \equiv 0$ which is increasing on $[0, \infty[$. It will be called a weight function if it has the following properties:

(α) there exists $K \geq 1$ with $\omega(2t) \leq K(1 + \omega(t))$ for all $t \geq 0$,

(β) $\int_1^\infty \frac{\omega(t)}{1+t^2} < \infty$,

(γ) $\lim_{t \rightarrow \infty} \frac{\log(1+t)}{\omega(t)} = 0$,

(δ) $\varphi: [0, \infty[\rightarrow [0, \infty[$, $\varphi(t) := \omega(e^t)$ is convex.

For $z \in \mathbb{C}^N$ we write $\omega(z)$ for $\omega(|z|)$, where $|z| = \sum_{j=1}^N |z_j|$.

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1.2 Lemma. *Let ω be as in 1.1 and assume that ω satisfies 1.1(α). Then we have for all $x, y \in \mathbb{T}^N$:*

$$\omega(x + y) \leq K(1 + \omega(x) + \omega(y)).$$

PROOF. By the properties of ω we have

$$\begin{aligned} \omega(x + y) &= \omega(|x + y|) \leq \omega(|x| + |y|) \leq \omega(2 \max(|x|, |y|)) \\ &\leq K(1 + \omega(\max(|x|, |y|))) \leq K(1 + \omega(x) + \omega(y)). \end{aligned}$$

The following elementary statements are well known and easy to check.

1.3 Remark. Let $\varphi: [0, \infty[\rightarrow [0, \infty[$ be convex and increasing with $\varphi(0) = 0$ and $\lim_{x \rightarrow \infty} x/\varphi(x) = 0$. Then we can define the Young conjugate φ^* of φ by

$$\varphi^*: [0, \infty[\rightarrow [0, \infty[, \quad \varphi^*(y) := \sup_{x \geq 0} (xy - \varphi(x)).$$

φ^* is convex and increasing and satisfies $\varphi^*(0) = 0$ and $\lim_{y \rightarrow \infty} y/\varphi^*(y) = 0$. Moreover, we have $(\varphi^*)^* = \varphi$.

1.4 Lemma. *For φ as in 1.3 assume that for some $L > 0$ we have $\varphi(x+1) \leq L(1+\varphi(x))$ for all $x \in [0, \infty[$. Then there exists $y_0 > 0$ so that*

$$\varphi^*(y) - y \geq L\varphi^*\left(\frac{y}{L}\right) - L \quad \text{for all } y \geq y_0.$$

1.5 Lemma. *Let $f: [0, \infty[\rightarrow [0, \infty[$ be convex and increasing with $f(0) = 0$. Then the function $h: x \mapsto f(x)/x$ is increasing on $[0, \infty[$.*

1.6 Lemma. *Let ω be as in 1.1 and assume that ω satisfies 1.1(α), (β), and $\log(1+t) = O(\omega(t))$ as t tends to ∞ . Then there exists a function $\sigma: [0, \infty[\rightarrow [0, \infty[$, satisfying 1.1(α)-(γ) with $\omega(t) = o(\sigma(t))$. If ω is a weight function, then σ can be chosen to be a weight function.*

PROOF. Determine inductively a sequence $(x_n)_{n \in \mathbb{N}}$ in $[0, \infty[$ with $x_1 = 0$ and $\omega(x_2) > 0$ which has the following properties for all $n \in \mathbb{N}$:

$$\int_{x_{n+1}}^{\infty} \frac{\omega(t)}{1+t^2} dt \leq \frac{1}{(n+1)^3}, \quad (1)$$

$$x_{n+1} \geq 2x_n, \quad (2)$$

$$\omega(x_{n+1}) \geq 2^{n+1-i} \omega(x_i) \quad \text{for } 1 \leq i \leq n. \quad (3)$$

Then define $\sigma: [0, \infty[\rightarrow [0, \infty[$ by

$$\sigma(x) := n\omega(x) - \sum_{i=1}^n \omega(x_i) \quad \text{for } |x| \in [x_n, x_{n+1}[. \quad (4)$$

It is easy to check that σ is a continuous even function. It has property 1.1(δ), provided ω has it. To show that $\omega(x) = o(\sigma(x))$, note that for $x \in [x_n, x_{n+1}[$, $n \geq 2$, we get from (3) and (4)

$$\sigma(x) = \left(n - \sum_{i=1}^n \frac{\omega(x_i)}{\omega(x)}\right)\omega(x) \geq \left(n - \sum_{i=1}^n 2^{i-n}\right)\omega(x) \geq (n-2)\omega(x),$$

which implies

$$\omega(x) \leq \frac{1}{n-2}\sigma(x) \quad \text{for } x \in [x_n, x_{n+1}[, n > 2. \quad (5)$$

Hence we have $\omega(x) = o(\sigma(x))$ and consequently σ satisfies 1.1(γ). To show that σ satisfies 1.1(α), note that 1.1(α) for ω implies $L := \sup_{x > x_3} \omega(2x)/\omega(x) < \infty$. Moreover, (2) implies that for $x > x_3$ we only have the following two cases:

CASE 1: $x_n \leq x < 2x < x_{n+1}$. Then (4), (5), and the definition of L imply

$$\sigma(2x) - \sigma(x) = n(\omega(2x) - \omega(x)) \leq n(L-1)\omega(x) \leq \frac{n}{n-2}(L-1)\sigma(x).$$

CASE 2: $x_n \leq x < x_{n+1} \leq 2x < x_{n+2}$. Then we have

$$\begin{aligned} \sigma(2x) - \sigma(x) &= (n+1)\omega(2x) - n\omega(x) - \omega(x_{n+1}) \\ &\leq (n+1)\omega(2x) - (n+1)\omega(x) \leq \frac{n+1}{n-2}(L-1)\sigma(x). \end{aligned}$$

Hence σ satisfies 1.1(α). Finally we get from (1) and (4)

$$\int_0^\infty \frac{\sigma(t)}{1+t^2} dt = \sum_{n=1}^\infty \int_{x_n}^{x_{n+1}} \frac{\sigma(t)}{1+t^2} dt \leq \sum_{n=1}^\infty \int_{x_n}^\infty \frac{n\omega(t)}{1+t^2} dt \leq \sum_{n=1}^\infty n \frac{1}{n^3} < \infty,$$

which shows that σ satisfies 1.1(β).

Remark. Clearly, Lemma 1.6 is inspired by Lemma 1 in section 3 of Beurling [1] (see Björck [2], 1.2.7). However, the proof is a lot easier since we replaced the subadditivity of ω by condition 1.1(α).

1.7 Lemma. *Let ω be a weight function and assume that $g: [0, \infty[\rightarrow [0, \infty[$ satisfies $g(t) = o(\omega(t))$ as t tends to ∞ . Then there exists a weight function σ with the following properties:*

- (i) $g(t) = o(\sigma(t))$ as $t \rightarrow \infty$,
- (ii) $\sigma(t) = o(\omega(t))$ as $t \rightarrow \infty$,
- (iii) for every $A > 1$: $\limsup_{t \rightarrow \infty} \frac{\sigma(At)}{\sigma(t)} \leq \limsup_{t \rightarrow \infty} \frac{\omega(At)}{\omega(t)}$.

If in addition there is $R \geq 1$ such that $\omega|_{[R, \infty[}$ is concave, then it is possible to make $\sigma|_{[R, \infty[}$ concave, too.

PROOF. We show first that if ω is concave on $[R, \infty[$, then $\omega \in C^1(]R, \infty[)$. For concave ω we have, for every $t > R$, left and right derivatives $\omega'_-(t)$ and $\omega'_+(t)$ with $\omega'_+(t) \leq \omega'_-(t)$. On the other hand, $\varphi(x) = \omega(e^x)$ is convex, hence $\omega'_-(t) = \varphi'_-(\log t)/t \leq \varphi'_+(\log t)/t = \omega'_+(t)$. Thus ω and φ are differentiable on $]R, \infty[$ and, by monotony of the derivative, even C^1 .

If in the general case ω is not C^1 , we choose a C^1 -function $\chi \geq 0$ with $\text{Supp } \chi \subset]0, \log 2[$ and $\int \chi = 1$ and define

$$\psi(x) = \int_{\mathbb{R}} \varphi(s+x)\chi(s)ds \quad \text{and} \quad \rho(t) = \psi(\log t), \quad t \geq 1.$$

For the constant K from 1.1(α) we have $\varphi \leq \psi \leq K\varphi$, which implies that ρ satisfies 1.1(β) and (γ). It is not difficult to see that ψ is convex and that, therefore, 1.1(δ) holds for ρ . Let $A, B > 1$ with $\omega(At) \leq B\omega(t)$ for large t be given. Set $a = \log A$. Then $\psi(x+a) = \int \varphi(s+x+a)\chi(s)ds \leq \int B\varphi(x+s)\chi(s)ds = B\varphi(x)$. Thus ρ satisfies 1.1(α) and $\limsup_{t \rightarrow \infty} \rho(At)/\rho(t) \leq \limsup_{t \rightarrow \infty} \omega(At)/\omega(t)$.

So far we have shown that we may assume that $\omega|]R, \infty[$ is C^1 for some R . We may also assume that $\log(1+t) = o(g(t))$. Remember that $\varphi(0) = 0$. We put $x_1 = y_1 = z_1 = 0$ and define inductively x_n, y_n, z_n with $x_2 > R$ and

$$x_n > y_{n-1} + n, \quad (1)$$

$$g(e^x) \leq \frac{\varphi(x)}{n^2} \quad \text{for all } x \geq x_n, \quad (2)$$

$$\varphi(x_n) \geq n \sum_{i=1}^{n-1} \varphi(z_i), \quad (3)$$

$$\varphi'(y_n) = \frac{n}{n-1} \varphi'(x_n), \quad (4)$$

$$\varphi(z_n) = n\varphi(x_n) - (n-1)\varphi(y_n) + n(y_n - x_n)\varphi'(x_n). \quad (5)$$

We claim

$$x_n \leq z_n \leq y_n. \quad (6)$$

From (5) and (4) we get

$$\begin{aligned} \frac{\varphi(z_n) - \varphi(x_n)}{y_n - x_n} &= -(n-1) \frac{\varphi(y_n) - \varphi(x_n)}{y_n - x_n} + n\varphi'(x_n) \\ &= (n-1) \left(\varphi'(y_n) - \frac{\varphi(y_n) - \varphi(x_n)}{y_n - x_n} \right), \end{aligned}$$

which is nonnegative because φ' is nondecreasing. On the other hand, (5) also implies

$$\frac{\varphi(y_n) - \varphi(z_n)}{y_n - x_n} = n \frac{\varphi(y_n) - \varphi(x_n)}{y_n - x_n} - n\varphi'(x_n) \geq 0.$$

We define ψ by

$$\psi(x) = \begin{cases} \frac{1}{n-1} \varphi(x_n) + \sum_{i=1}^{n-2} \frac{1}{i(i+1)} \varphi(z_{i+1}) + \frac{x - x_n}{n-1} \varphi'(x_n), & x_n \leq x < y_n, \\ \frac{1}{n} \varphi(x) + \sum_{i=1}^{n-1} \frac{1}{i(i+1)} \varphi(z_{i+1}), & y_n \leq x < x_{n+1}. \end{cases}$$

It is a consequence of (4) and (5) that ψ is C^1 . Since it consists only of linear parts and of dilated and shifted parts of φ , it is convex. We define σ by

$$\sigma(t) = \psi(\max(\log t, 0)).$$

This is again a C^1 -function. The argument that shows the convexity of ψ also yields that σ is concave provided ω is concave.

We claim

$$\psi(x) \geq \frac{1}{n}\varphi(x) \quad \text{for } x \in [x_n, x_{n+1}], \quad n \geq 2. \quad (7)$$

This is obvious for $x \in [y_n, x_{n+1}]$. For $x \in [x_n, y_n]$ we get from (4) and the convexity of φ

$$\psi(x) \geq \frac{1}{n-1}\varphi(x_n) + \frac{x-x_n}{n-1}\varphi'(x_n) \geq \frac{1}{n}\varphi(x_n) + \frac{x-x_n}{n}\varphi'(y_n) \geq \frac{1}{n}\varphi(x).$$

Because of (2), the estimate (7) proves (i).

For the proof of (ii), we first consider the case $y_n \leq x \leq x_{n+1}$. We apply (3) and (6):

$$\frac{\psi(x)}{\varphi(x)} = \frac{1}{n} + \sum_{i=1}^{n-1} \frac{1}{i(i+1)} \frac{\varphi(z_{i+1})}{\varphi(x)} \leq \frac{1}{n} + \frac{1}{n(n-1)} + \frac{1}{n}.$$

Now let $x_n \leq x \leq y_n$. Again we use the convexity of φ :

$$\begin{aligned} \frac{\psi(x)}{\varphi(x)} &= \frac{1}{n-1} \frac{\varphi(x_n)}{\varphi(x)} + \sum_{i=1}^{n-2} \frac{1}{i(i+1)} \frac{\varphi(z_{i+1})}{\varphi(x)} + \frac{x-x_n}{n-1} \frac{\varphi'(x_n)}{\varphi(x)} \\ &\leq \frac{1}{n-1} \frac{\varphi(x_n)}{\varphi(x)} + \frac{1}{n-1} (x-x_n) \frac{\varphi'(x_n)}{\varphi(x)} + \frac{1}{n} \leq \frac{1}{n-1} \frac{\varphi(x)}{\varphi(x)} + \frac{1}{n}. \end{aligned}$$

To prove (iii), fix $A > 1$, set $a = \log A$, and estimate $\psi(x+a) - \psi(x)$ in terms of $\psi(x)$. Choose $B > \limsup \omega(At)/\omega(t)$, and choose $n > a$ so large that for all $x \geq x_n$ we have $\varphi(x+a) \leq B\varphi(x)$. Now we have to look at several cases, depending into which intervals x and $x+a$ fall. Because of (1) and $n > a$ there are only finitely many cases—five to be precise. We only discuss two of them, the others are similar.

CASE $x_n \leq x \leq y_n \leq x+a \leq x_{n+1}$: We apply (5) in the first step, then we use the estimate $\varphi'(x) \leq (\varphi(y_n) - \varphi(x))/(y_n - x)$, and we refer to (7) at the end.

$$\begin{aligned} \psi(x+a) - \psi(x) &= \frac{1}{n}\varphi(x+a) - \frac{1}{n}\varphi(y_n) + \frac{y_n-x}{n-1}\varphi'(x_n) \\ &\leq \frac{1}{n}\varphi(x+a) - \frac{1}{n}\varphi(y_n) + \frac{1}{n-1}(\varphi(y_n) - \varphi(x)) \\ &\leq \frac{1}{n-1}(\varphi(x+a) - \varphi(x)) \leq \frac{B-1}{n-1}\varphi(x) \leq (B-1)\frac{n}{n-1}\psi(x). \end{aligned}$$

CASE $x_n \leq x < x + a \leq y_n$: Again we use (7) in the last step:

$$\psi(x + a) - \psi(x) = \frac{a}{n-1} \varphi'(x_n) \leq \frac{a}{n-1} \frac{\varphi(x_n + a) - \varphi(x_n)}{a} \leq \frac{B-1}{n-1} \varphi(x_n) \leq (B-1)\psi(x).$$

In particular, (iii) implies 1.1(α), while (ii) implies 1.1(β). Since we assumed $\log = o(g)$, we can derive 1.1(γ) from (i).

1.8 Remark. (1) If ω is a weight function except that it does not satisfy 1.1(β), then the function σ can be constructed as above, except that it may not satisfy 1.1(β).

(2) Note that the proof shows that σ can be constructed so that it coincides with ω on a given arbitrarily large bounded interval.

(3) Using (2), it is easy to see the following fact, which is used in Meise and Taylor [22]:

If ω and ρ are weight functions, concave on $]R, \infty[$, with $\rho = o(\omega)$, then for each $D > 0$ there is a weight function σ as in the lemma with the additional property that $\sigma \equiv \rho$ on $[0, D]$.

In fact, because of $\lim_{t \rightarrow \infty} \omega'(t) = 0$ there is $E > 0$ with $\omega'(E) = \rho'(D)$ provided D is sufficiently large. We define

$$\tau(t) = \begin{cases} \rho(t) & \text{if } t \leq D, \\ \omega(t - D + E) + \rho(D) - \omega(E), & \text{if } t \geq D. \end{cases}$$

Since τ is C^1 , and ρ and ω are concave weight functions, also τ is concave and $x \mapsto \tau(e^x)$ is convex. The other axioms for weight functions are inherited from ω . To get σ , we apply Lemma 1.7 with τ in the rôle of ω and ρ in the rôle of g requiring $g \equiv \tau$ on $[0, D]$.

1.9 Proposition. *Let ω be a weight function, and for $j \in \mathbb{N}$ let $g_j: [0, \infty[\rightarrow [0, \infty[$ satisfy $g_j(t) = o(\omega(t))$ as t tends to ∞ . Then there exists a weight function σ with the following properties*

- (i) $\lim_{t \rightarrow \infty} \frac{g_j(t)}{\sigma(t)} = 0$ for each $j \in \mathbb{N}$,
- (ii) $\lim_{t \rightarrow \infty} \frac{\sigma(t)}{\omega(t)} = 0$,
- (iii) for every $A > 1$: $\limsup_{t \rightarrow \infty} \frac{\sigma(At)}{\sigma(t)} \leq \limsup_{t \rightarrow \infty} \frac{\omega(At)}{\omega(t)}$.

PROOF. Choose an increasing sequence $(x_j)_{j \in \mathbb{N}}$ in $[0, \infty[$ which satisfies

$$\max_{1 \leq i \leq j} g_i(x) \leq \omega(x)/j \quad \text{for all } x \in [0, \infty[\text{ with } x \geq x_j.$$

Then define $g: [0, \infty[\rightarrow [0, \infty[$ by $g|] - x_1, x_1[\equiv 0$ and $g(x) = \max_{1 \leq i \leq j} g_i(x)$ if $x_j \leq x < x_{j+1}$, $j \in \mathbb{N}$. By the definition of the sequence $(x_j)_{j \in \mathbb{N}}$ we have $g(t) = o(\omega(t))$ as t tends to ∞ . Hence the result follows from Lemma 1.7.

2 Construction of functions with compact support

In this section we use a modification of an idea of Beurling [1] (see also Björck [2], 1.3.12) to prove the existence of continuous functions with compact support for which the Fourier transform decays very rapidly.

2.1 Definition. Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be as in 1.1 and assume that ω satisfies 1.1(α) and 1.1(β). Let $H := \{x + iy \mid x \in \mathbb{R}, y > 0\}$ denote the upper half plane and define the harmonic extension $P_\omega: H \rightarrow [0, \infty[$ by

$$P_\omega(x + iy) = \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{\omega(t)}{(t-x)^2 + y^2} dt.$$

2.2 Lemma. For ω as in 2.1 the harmonic extension P_ω is harmonic on H and satisfies the following estimates

- (1) $P_\omega(x + iy) \geq \frac{1}{4}\omega(|x + iy|) \geq \frac{1}{4}\omega(x)$ for all $x + iy \in H$,
- (2) for each $\varepsilon > 0$ there exists $A > 0$ with $P_\omega(x + iy) \leq \varepsilon y + K\omega(x) + A$ for all $x + iy \in H$.

(In fact, $P_\omega \geq \omega$ holds. However, (1) is easier to show and sufficient for our purpose.)

PROOF. It is well known that P_ω is harmonic in H . Because of the symmetry properties of ω and of the Poisson kernel, it suffices to prove (1) for $x \geq 0$. Since $\omega|_{[0, \infty[}$ is increasing, we have

$$\begin{aligned} P_\omega(x + iy) &\geq \frac{y}{\pi} \int_{x+y}^{\infty} \frac{\omega(t)}{(t-x)^2 + y^2} dt = \frac{y}{\pi} \int_y^{\infty} \frac{\omega(x+t)}{t^2 + y^2} dt \geq \frac{y}{\pi} \int_y^{\infty} \frac{\omega(x+y)}{t^2 + y^2} dt \\ &= \omega(x+y) \frac{1}{\pi} \int_y^{\infty} \frac{y}{t^2 + y^2} dt = \frac{1}{4}\omega(x+y) \geq \frac{1}{4}\omega(|x + iy|) \geq \frac{1}{4}\omega(x). \end{aligned}$$

To prove (2), let $\varepsilon > 0$ be given. Then 1.1(β) implies the existence of $R = R(\varepsilon) \geq 1$ with

$$\frac{2K}{\pi} \int_R^{\infty} \frac{\omega(t)}{t^2} dt \leq \varepsilon,$$

where K denotes the constant in 1.1(α). Then 1.2 implies for each $x + iy \in H$

$$\begin{aligned} P_\omega(x + iy) &= \frac{y}{\pi} \int_{-\infty}^{+\infty} \frac{\omega(t+x)}{t^2 + y^2} dt \leq K(1 + \omega(x)) + 2K \frac{y}{\pi} \int_0^{\infty} \frac{\omega(t)}{t^2 + y^2} dt \\ &= K(1 + \omega(x)) + 2K \frac{y}{\pi} \left(\int_0^R \frac{\omega(t)}{t^2 + y^2} dt + \int_R^{\infty} \frac{\omega(t)}{t^2 + y^2} dt \right) \\ &\leq K(1 + \omega(x)) + K\omega(R) + 2K \frac{y}{\pi} \int_R^{\infty} \frac{\omega(t)}{t^2} dt \leq \varepsilon y + K\omega(x) + A, \end{aligned}$$

where $A = K(1 + \omega(R(\varepsilon)))$.

2.3 Lemma. *Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be as in 1.1 and assume that ω satisfies 1.1(α)-(γ). Then there exists a nonzero function $g \in \mathcal{S}(\mathbb{R})$ with support in $] -\infty, 0]$ for which the Fourier transform $\hat{g}$ satisfies*

$$|\hat{g}(x)| \leq \exp(-2K\omega(x)) \quad \text{for all } x \in \mathbb{R}, \quad (1)$$

where K denotes the constant in 1.1(α).

PROOF. By Lemma 2.2 the function $u := P_\omega$ is harmonic in the upper half plane H . Let v denote a harmonic conjugate of u , so that $h := u + iv$ is holomorphic in H . Then put $L := 8K$, define $F: H \rightarrow \mathbb{C}$ by $F := \exp(-Lh)$ and define $f: \mathbb{R} \rightarrow \mathbb{C}$ by

$$f(x) := F(x + i).$$

From 2.2(1) we get for all $x + iy \in H$

$$|F(x + iy)| = \exp(-LP_\omega(x + iy)) \leq \exp(-2K\omega(x)). \quad (2)$$

Since F is holomorphic on H we have for each $j \in \mathbb{N}$ and $x \in \mathbb{R}$

$$f^{(j)}(x) = \frac{d^j F}{dz^j}(x + i) = \frac{j!}{2\pi i} \int_{|x+i-\zeta|=1/2} \frac{F(\zeta)}{(x+i-\zeta)^{j+1}} d\zeta. \quad (3)$$

The properties of ω imply the existence of some $B > 0$ such that for all $x \in \mathbb{R}$ we have

$$\omega(x) \leq K\omega(|x| - 1/2) + B.$$

Using this and (2), we get from (3) that for each $j \in \mathbb{N}$ there exists $C_j > 0$ with

$$|f^{(j)}(x)| \leq C_j \exp(-2\omega(x)) \quad \text{for all } x \in \mathbb{R}. \quad (4)$$

Since ω satisfies 1.1(γ), it follows from (4) and (2) that f belongs to the space $\mathcal{S}(\mathbb{R})$ of all rapidly decreasing C^∞ -functions. Hence there exists $g \in \mathcal{S}(\mathbb{R})$ with

$$f = \hat{g}: x \mapsto \int_{-\infty}^{+\infty} g(t) e^{-ixt} dt, \quad (5)$$

since the Fourier transform is an isomorphism of $\mathcal{S}(\mathbb{R})$. By definition, F is not identically zero in H . Hence f and, consequently, g is not the zero function. From (2) and (5) we get (1).

To show $\text{Supp}(g) \subset] -\infty, 0]$ we note that the estimate (2), Cauchy's integral theorem, and Fourier's inversion formula imply by standard arguments that we have for each $x > 0$

$$g(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(t + iy) \exp(ix(t + i(y - 1))) dt \quad \text{for each } y \geq 1. \quad (6)$$

Estimating (6) by (2) and taking the limit for y tending to ∞ , we get $g(x) = 0$ for all $x > 0$.

2.4 Proposition. *Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be as in 1.1 and assume that ω satisfies 1.1(α)-(γ). Then for each $\varepsilon > 0$ there exists $h \in C^\infty(\mathbb{R})$, $h \neq 0$, with*

$$\text{Supp}(h) \subset [-\varepsilon, \varepsilon], \quad (1)$$

$$\int_{-\infty}^{+\infty} |\hat{h}(x)| \exp(\omega(x)) dx < \infty. \quad (2)$$

PROOF. Let g be the function which was constructed in Lemma 2.3. Because of well known properties of the Fourier transform, we can assume $0 \in \text{Supp}(g)$. Hence, for fixed $\varepsilon > 0$, we find $x_0 < 0$ with

$$-\varepsilon < x_0 < 0 \quad \text{and} \quad g(x_0) \neq 0. \quad (3)$$

Then the function $h: \mathbb{R} \rightarrow \mathbb{C}$,

$$h(x) := g(x_0 + x)g(x_0 - x)$$

is in $C^\infty(\mathbb{R})$ and satisfies (1). Because of our choice of x_0 we have $h(0) \neq 0$.

To show (2), we put $f := \hat{g}$. Then it is easy to check that for all $x \in \mathbb{R}$

$$\hat{h}(t) = e^{ix_0 t} \int_{-\infty}^{+\infty} f(t-s)f(-s)e^{-2ix_0 s} ds.$$

From this we get by 2.3(1), 1.2, and 1.1(γ)

$$\begin{aligned} \int_{-\infty}^{+\infty} |\hat{h}(t)| e^{\omega(t)} dt &\leq \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \exp(-2K(\omega(t-s) + \omega(-s)) + \omega(t)) ds dt \\ &\leq \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \exp(-2K(\omega(t-s) + \omega(-s)) + K(\omega(t-s) + \omega(s) + 1)) ds dt \\ &\leq e^K \left(\int_{-\infty}^{+\infty} \exp(-K\omega(t)) dt \right)^2 < \infty. \end{aligned}$$

2.5 Corollary. *Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be as in 1.1 and assume that ω satisfies 1.1(α)-(γ). Then for each $N \in \mathbb{N}$ there exists $\delta_N > 0$ such that for each $\varepsilon > 0$ there exists $H \in C^\infty(\mathbb{R}^N)$, $H \neq 0$, with*

$$\text{Supp}(H) \subset [-\varepsilon, \varepsilon]^N, \quad (1)$$

$$\int_{\mathbb{R}^N} |\widehat{H}(t)| \exp(\delta_N \omega(t)) dt < \infty. \quad (2)$$

PROOF. The properties of ω imply that for each $N \in \mathbb{N}$ there exist $\delta_N > 0$ and $C_N > 0$ such that for all $t \in \mathbb{R}^N$

$$\delta_N \omega(t) \leq \sum_{j=1}^N \omega(t_j) + C_N. \quad (3)$$

If $\varepsilon > 0$ is given, we choose $h \in C^\infty(\mathbb{R})$ according to Proposition 2.4 and define

$$H: (x_1, \dots, x_N) \mapsto \prod_{j=1}^N h(x_j).$$

Then H is in $C^\infty(\mathbb{R}^N)$ and satisfies (1). Since $\widehat{H}(t_1, \dots, t_N) = \prod_{j=1}^N \hat{h}(t_j)$, condition (2) follows easily from (3) and 2.4(2).

2.6 Corollary. *Let $\omega: [0, \infty[\rightarrow [0, \infty[$ be as in 1.1 and assume that ω satisfies 1.1(α)-(γ). Then for each $N \in \mathbb{N}$ and each $\varepsilon > 0$ there exists $H \in C^\infty(\mathbb{R}^N)$, $H \neq 0$, with*

$$\text{Supp}(H) \subset [-\varepsilon, \varepsilon]^N, \quad (1)$$

$$\int_{\mathbb{R}^N} |\widehat{H}(t)| \exp(m\omega(t)) dt < \infty \quad \text{for all } m > 0. \quad (2)$$

PROOF. By 1.6 we can choose a function $\sigma: [0, \infty[\rightarrow [0, \infty[$ as in 1.1 which satisfies 1.1(α)-(γ) as well as $\lim_{t \rightarrow \infty} \omega(t)/\sigma(t) = 0$. Now fix $N \in \mathbb{N}$ and $\varepsilon > 0$. Then choose $H \in C^\infty(\mathbb{R}^N)$ such that 2.5(1) and (2) hold for ω replaced by σ . Then it follows easily from 2.5(2) for σ and $\lim_{t \rightarrow \infty} \omega(t)/\sigma(t) = 0$ that H satisfies also condition (2).

Remark. The conclusions of Corollaries 2.5 and 2.6 hold for all functions $\omega: [0, \infty[\rightarrow [0, \infty[$ which are increasing and continuous and satisfy condition 1.1(β). This follows from a result of Ciorănescu and Zsidó [9] and these Corollaries, as we show in 8.7.

3 Ultradifferentiable functions with compact support

In this section we define spaces of test functions and prove their basic properties and a Paley-Wiener theorem following Beurling [1] and Petzsche and Vogt [24].

3.1 Definition. Let ω as in 1.1 satisfy 1.1(α)-(γ) and let $K \subset \mathbb{R}^N$ be a compact set.

(1) For $\lambda > 0$ we define the Banach space

$$\mathcal{D}_\lambda(K) = \left\{ f \in C^\infty(\mathbb{R}^N) \mid \text{Supp}(f) \subset K \quad \text{and} \right. \\ \left. \|f\|_\lambda := \int_{\mathbb{R}^N} |\widehat{f}(t)| \exp(\lambda\omega(t)) dt < \infty \right\}.$$

(2) We set

$$\mathcal{D}_{\{\omega\}}(K) = \text{ind}_{\lambda \rightarrow 0} \mathcal{D}_\lambda(K)$$

endowed with the topology of the inductive limit, and we set

$$\mathcal{D}_{(\omega)}(K) = \text{proj}_{\lambda \rightarrow \infty} \mathcal{D}_\lambda(K)$$

endowed with the topology of the projective limit.

(3) For an open set $\Omega \subset \mathbb{R}^N$ we define

$$\mathcal{D}_{\{\omega\}}(\Omega) = \text{ind}_K \mathcal{D}_{\{\omega\}}(K) \quad \text{and} \quad \mathcal{D}_{(\omega)}(\Omega) = \text{ind}_K \mathcal{D}_{(\omega)}(K)$$

where the inductive limits are taken over all compact subsets of Ω . $\mathcal{D}_{\{\omega\}}(\Omega)$ and $\mathcal{D}_{(\omega)}(\Omega)$ carry the respective inductive limit topologies.

The elements of $\mathcal{D}_{(\omega)}(\Omega)$ (resp. $\mathcal{D}_{\{\omega\}}(\Omega)$) are called test functions of Beurling type (resp. of Roumieu) type.

3.2 Remark. (1) For ω as in 3.1 and $K \subset \mathbb{R}^N$ with nonempty interior we have $\{0\} \neq \mathcal{D}_{(\omega)}(K) \subset \mathcal{D}_{\{\omega\}}(K)$. The second inclusion is obvious from the definitions. To see the first part of our claim assume that $0 \in \overset{\circ}{K}$, choose ε with $[-\varepsilon, \varepsilon]^N \subset K$, and apply 2.6.

(2) For ω and σ as in 1.1 with properties (α) – (γ) , Franken [10], 1.13, has shown that $\mathcal{D}_{(\omega)}(\mathbb{R}) \subset \mathcal{D}_{(\sigma)}(\mathbb{R})$ if and only if $\sigma = O(\omega)$ (see also Björck [2], 1.3.18).

(3) We say that functions ω and σ are equivalent if $\omega = O(\sigma)$ and $\sigma = O(\omega)$. Note that if $\omega \leq \sigma \leq C\omega$ for some $C > 0$ and if $\psi(x) = \sigma(e^x)$, then

$$C\varphi^*(y/C) \leq \psi^*(y) \leq \varphi^*(y) \quad \text{for all } y > 0.$$

With this formula, it is easy to see that the definitions and most theorems in the sequel do not change if ω is only equivalent to a weight function.

Remember that for compact $K \subset \mathbb{R}^N$ the support function is defined as $H_K(x) = \sup_{y \in K} \langle x, y \rangle$.

3.3 Lemma. *Let ω be a weight function and let $f \in \mathcal{D}(\mathbb{R}^N)$.*

(a) *If there is $B > 0$ with*

$$\int_{\mathbb{R}^N} |\hat{f}(t)| e^{B\omega(t)} dt =: C < \infty$$

then we have

$$\sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in \mathbb{R}^N} |f^{(\alpha)}(x)| \exp \left(-B\varphi^* \left(\frac{|\alpha|}{B} \right) \right) \leq \frac{C}{(2\pi)^N}. \quad (1)$$

(b) *If (1) holds for f and B then there is $D > 0$, depending only on ω , N , and B , and $L > 0$, depending only on ω and N , such that with $K = \text{Supp } f$ and $m_N(K)$ its Lebesgue measure we have for all $z \in \mathbb{C}^N$:*

$$|\hat{f}(z)| \leq m_N(K) \frac{CD}{(2\pi)^N} \exp \left(H_K(\text{Im } z) - \frac{B}{L} \omega(z) \right). \quad (2)$$

PROOF. 1.) From $f^{(\alpha)}(x) = (2\pi)^{-N} \int_{\mathbb{R}^N} \hat{f}(t) (it)^\alpha \exp(i\langle x, t \rangle) dt$ we derive the following estimates:

$$\begin{aligned} |f^{(\alpha)}(x)| &\leq \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} |\hat{f}(t)| e^{B\omega(t)} \exp \left(-B\omega(t) + \sum_{j=1}^N \alpha_j \log |t_j| \right) dt \\ &\leq \frac{C}{(2\pi)^N} \sup_{t \in \mathbb{R}^N} \exp (|\alpha| \log |t| - B\omega(t)) \\ &\leq \frac{C}{(2\pi)^N} \sup_{y > 0} \exp (|\alpha| y - B\varphi(y)) \\ &= \frac{C}{(2\pi)^N} \exp \left(B \sup_{y > 0} \left(\frac{|\alpha|}{B} y - \varphi(y) \right) \right) = \frac{C}{(2\pi)^N} \exp \left(B\varphi^* \left(\frac{|\alpha|}{B} \right) \right). \end{aligned}$$

2.) By 1.1(α) there is $L > 0$ such that $\omega(Nr) \leq (L/2)\omega(r) + L/2$ for all $r > 0$. By 1.1(γ) there is $D' > 0$ such that $(2B/L)\omega(r) - \log r \geq (B/L)\omega(r) - D'$ for all $r > 1$.

Let now $z \in \mathbb{C}^N$ be given, let l be the index with $|z_l| = \max_{j=1}^N |z_j|$, and assume $|z_l| > 1$. Then we have by partial integration for all $j \in \mathbb{N}_0$:

$$\hat{f}(z) = \int_K \frac{\partial^j}{\partial t_l^j} f(t) \frac{1}{(iz_l)^j} e^{-i\langle t, z \rangle} dt.$$

In view of (1) this implies for all $j \in \mathbb{N}_0$:

$$|\hat{f}(z)| \leq m_N(K) \frac{C}{(2\pi)^N} \exp \left(B\varphi^* \left(\frac{j}{B} \right) - j \log |z_l| + H_K(\operatorname{Im} z) \right). \quad (3)$$

Note that

$$\begin{aligned} \sup_{j \in \mathbb{N}_0} \left(j \log |z_l| - B\varphi^* \left(\frac{j}{B} \right) \right) &= B \sup_{j \in \mathbb{N}_0} \left(\frac{j+1}{B} \log |z_l| - \varphi^* \left(\frac{j}{B} \right) \right) - \log |z_l| \\ &\geq B \sup_{x>0} (x \log |z_l| - \varphi^*(x)) - \log |z_l| = B\varphi^{**}(\log |z_l|) - \log |z_l| \\ &= B\omega(z_l) - \log |z_l| \geq B\omega \left(\frac{z}{N} \right) - \log |z| \\ &\geq \frac{2B}{L}\omega(z) - 1 - \log |z| \geq \frac{B}{L}\omega(z) - D'. \end{aligned}$$

By passing to the infimum over all $j \in \mathbb{N}_0$ in (3) we see that for all $z \in \mathbb{C}^N$ with $\max_{j=1}^N |z_j| > 1$ we have:

$$|\hat{f}(z)| \leq m_N(K) \frac{C}{(2\pi)^N} \exp \left(H_K(\operatorname{Im} z) - \frac{B}{L}\omega(z) + D' \right).$$

If we let $D = \exp(D' + (B/L)\omega(N))$ we see from (3) applied to $j = 0$ that the claim also holds if $\max_{j=1}^N |z_j| \leq 1$.

Lemma 3.3 and the Paley-Wiener theorem for $\mathcal{D}(K)$ (Hörmander [13], 7.3.1) imply the following proposition:

3.4 Proposition. *Let ω be a weight function, $K \subset \mathbb{R}^N$ a convex compact set, and let $f \in L^1(\mathbb{R}^N)$ be given.*

(1) *The following are equivalent:*

- (a) $f \in \mathcal{D}_{\{\omega\}}(K)$,
- (b) $f \in \mathcal{D}(K)$ and for some $k \in \mathbb{N}$

$$\sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in \mathbb{R}^N} |f^{(\alpha)}(x)| \exp \left(-\frac{1}{k} \varphi^*(|\alpha|k) \right) < \infty,$$

(c) there are $\varepsilon, C > 0$ such that for all $z \in \mathbb{C}^N$

$$|\hat{f}(z)| \leq C \exp(H_K(\operatorname{Im} z) - \varepsilon \omega(z)).$$

(2) The following are equivalent:

(a) $f \in \mathcal{D}_{(\omega)}(K)$,

(b) $f \in \mathcal{D}(K)$ and for all $k \in \mathbb{N}$

$$\sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in \mathbb{R}^N} |f^{(\alpha)}(x)| \exp\left(-k\varphi^*\left(\frac{|\alpha|}{k}\right)\right) < \infty,$$

(c) for all $k \in \mathbb{N}$ there is $C > 0$ such that for all $z \in \mathbb{C}^N$

$$|\hat{f}(z)| \leq C \exp(H_K(\operatorname{Im} z) - k\omega(z)).$$

For a sequence $\mathbb{P} = (p_n)_{n \in \mathbb{N}}$ of continuous functions $p_n: \mathbb{C}^N \rightarrow \mathbb{R}$, we define

$$A_{\mathbb{P}}(\mathbb{C}^N) = \left\{ f \in H(\mathbb{C}^N) \mid \text{for some } n: \sup_{z \in \mathbb{C}^N} |f(z)| \exp(-p_n(z)) < \infty \right\},$$

and

$$A_{\mathbb{P}}^0(\mathbb{C}^N) = \left\{ f \in H(\mathbb{C}^N) \mid \text{for all } n: \sup_{z \in \mathbb{C}^N} |f(z)| \exp(-p_n(z)) < \infty \right\}.$$

The Paley-Wiener theorem 3.4 implies:

3.5 Proposition. *Let ω be a weight function.*

(1) Let $K \subset \mathbb{R}^N$ be a convex compact set. We set

$$\mathbb{P} = \left\{ p_n: z \mapsto H_K(\operatorname{Im} z) - \frac{1}{n}\omega(z) \mid n \in \mathbb{N} \right\}$$

and

$$\mathbb{M} = \left\{ m_n: z \mapsto H_K(\operatorname{Im} z) - n\omega(z) \mid n \in \mathbb{N} \right\}.$$

Then $\mathcal{D}_{\{\omega\}}(K) \cong A_{\mathbb{P}}(\mathbb{C}^N)$ and $\mathcal{D}_{(\omega)}(K) \cong A_{\mathbb{M}}^0(\mathbb{C}^N)$, where the isomorphisms are given by the Fourier-Laplace transform (see 3.3).

(2) Let $\Omega \subset \mathbb{R}^N$ be an open convex set. Let $\overset{\circ}{K}_1 \subset K_1 \subset \overset{\circ}{K}_2 \subset \dots$ be a convex compact exhaustion of Ω , and set

$$\mathbb{P} = \left\{ p_n: z \mapsto H_{K_n}(\operatorname{Im} z) - \frac{1}{n}\omega(z) \mid n \in \mathbb{N} \right\}.$$

Then $\mathcal{D}_{\{\omega\}}(\Omega) \cong A_{\mathbb{P}}(\mathbb{C}^N)$.

3.6 Corollary. *Let ω be a weight function, $K \subset \mathbb{R}^N$ be compact and $\Omega \subset \mathbb{R}^N$ be open.*

(1) $\mathcal{D}_{\{\omega\}}(K)$ is a (DFN)-space—i.e., the strong dual of a nuclear Fréchet space. In particular, it is complete, reflexive, and nuclear.

(2) $\mathcal{D}_{(\omega)}(K)$ is an (FN)-space—i.e., a nuclear Fréchet space.

PROOF. For convex K , the statements about $\mathcal{D}_{\{\omega\}}(K)$ and $\mathcal{D}_{(\omega)}(K)$ follow from 3.5. For arbitrary K , the space $\mathcal{D}_{(\omega)}(K)$ is a closed subspace of $\mathcal{D}_{(\omega)}(L)$ for some convex compact $L \supset K$, thus nuclear. This argument also works in the Roumieu case.

3.7 Lemma. *Let ω be a weight function, $f \in \mathcal{D}(\mathbb{R}^N)$, $g \in \mathcal{D}_{\{\omega\}}(\mathbb{R}^N)$. Then we have:*

$$f * g \in \mathcal{D}_{\{\omega\}}(\mathbb{R}^N) \quad (1)$$

$$\text{Supp}(f * g) \subset \text{Supp}(f) + \text{Supp}(g) \quad (2)$$

$$\widehat{f * g}(z) = \widehat{f}(z)\widehat{g}(z). \quad (3)$$

If $g \in \mathcal{D}_{(\omega)}(\mathbb{R}^N)$ then we even have

$$f * g \in \mathcal{D}_{(\omega)}(\mathbb{R}^N). \quad (4)$$

PROOF. Since $\mathcal{D}_{\{\omega\}}(\mathbb{R}^N) \subset \mathcal{D}(\mathbb{R}^N)$, (2) and (3) follow from the corresponding statements for ordinary test functions. So we only have to prove (1): Choose $A, B > 0$ such that $\text{Supp } f \subset [-A, A]^N$, $\text{Supp } g \subset [-B, B]^N$. Then there is $k \in \mathbb{N}$ such that with suitable $C, D > 0$ we have for all $z \in \mathbb{C}$:

$$\begin{aligned} |\widehat{f * g}(z)| &\leq |\widehat{f}(z)| |\widehat{g}(z)| \leq C \exp(A |\text{Im } z|) D \exp(B |\text{Im } z| - \omega(z)/k) \\ &\leq CD \exp((A + B) |\text{Im } z| - \omega(z)/k). \end{aligned}$$

Thus 3.4 yields our claim. The proof of (4) is strictly analogous.

3.8 Lemma. *Let $\sigma \leq \omega$ be two functions with properties 1.1(α)–(γ). Let $K_1, K_2 \subset \mathbb{R}^N$ be compact sets with $K_1 \subset \overset{\circ}{K}_2$. For each $f \in \mathcal{D}_{\{\sigma\}}(K_1)$ (resp. $\mathcal{D}_{(\sigma)}(K_1)$), there is a sequence $(f_n)_{n \in \mathbb{N}}$ in $\mathcal{D}_{\{\omega\}}(K_2)$ with $\lim_{n \rightarrow \infty} f_n = f$ in $\mathcal{D}_{\{\sigma\}}(K_2)$ (resp. $\mathcal{D}_{(\sigma)}(K_2)$). The analogous statement holds if $\mathcal{D}_{\{\sigma\}}$ is replaced by $\mathcal{D}$.*

PROOF. By 3.2 there is $\chi \in \mathcal{D}_{\{\omega\}}([-1, 1]^N)$ with $\int_{[-1, 1]^N} \chi(t) dt = 1$. Let $\chi_\varepsilon(t) = \varepsilon^{-N} \chi(t/\varepsilon)$, then $\text{Supp } \chi_\varepsilon \subset [-\varepsilon, \varepsilon]^N$. Lemma 3.7 implies that $f * \chi_\varepsilon \in \mathcal{D}_{\{\omega\}}(\mathbb{R}^N)$ with $\text{Supp}(f * \chi_\varepsilon) \subset K_1 + [-\varepsilon, \varepsilon]^N$. Thus for sufficiently small ε we have $f * \chi_\varepsilon \in \mathcal{D}_{\{\omega\}}(K_2)$. If $k \in \mathbb{N}$ is given with $\int_{\mathbb{R}^N} |\widehat{f}(t)| \exp(\sigma(t)/k) dt < \infty$, we apply 3.7 to get

$$\int_{\mathbb{R}^N} |\widehat{f}(t) - \widehat{f * \chi_\varepsilon}(t)| e^{(1/k)\sigma(t)} dt \leq \int_{\mathbb{R}^N} |\widehat{f}(t)| |1 - \chi(\varepsilon t)| e^{(1/k)\sigma(t)} dt.$$

Since $\lim_{\varepsilon \rightarrow 0} \widehat{\chi}(\varepsilon) = \widehat{\chi}(0) = 1$, we see that $\lim \|f - f * \chi_\varepsilon\|_{\sigma, k} = 0$, which proves our claim. The same proof works in the other cases.

3.9 Proposition. *Let ω and σ be weight functions, and let $\sigma = o(\omega)$. Then the inclusions*

$$\mathcal{D}_{(\omega)}(\Omega) \hookrightarrow \mathcal{D}_{\{\omega\}}(\Omega) \hookrightarrow \mathcal{D}_{(\sigma)}(\Omega) \hookrightarrow \mathcal{D}(\Omega)$$

are continuous and sequentially dense for each open $\Omega \subset \mathbb{R}^N$.

4 Ultradifferentiable functions with arbitrary support

In this section we introduce algebras of ω -ultradifferentiable functions of Beurling and of Roumieu type.

4.1 Definition. For a weight function ω and an open set $\Omega \subset \mathbb{R}^N$ we define

$$\begin{aligned} \mathcal{E}_{\{\omega\}}(\Omega) = \{ & f \in C^\infty(\Omega) \mid \text{for all compact } K \subset \Omega \text{ there is } m \in \mathbb{N} \\ & \text{with } \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K} |f^{(\alpha)}(x)| \exp\left(-\frac{1}{m} \varphi^*(m|\alpha|)\right) < \infty \} \end{aligned}$$

and

$$\begin{aligned} \mathcal{E}_{(\omega)}(\Omega) = \{ & f \in C^\infty(\Omega) \mid \text{for all compact } K \subset \Omega \text{ and all } m \in \mathbb{N} \\ & p_{K,m}(f) := \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K} |f^{(\alpha)}(x)| \exp\left(-m \varphi^*\left(\frac{|\alpha|}{m}\right)\right) < \infty \}. \end{aligned}$$

The topology of $\mathcal{E}_{\{\omega\}}(\Omega)$ is given by first taking the inductive limit over all $m \in \mathbb{N}$ for each compact $K \subset \Omega$ and then taking the projective limit of these, while $\mathcal{E}_{(\omega)}(\Omega)$ carries the metric locally convex topology given by the seminorms $p_{K,m}$ where K is a compact subset of Ω and $m \in \mathbb{N}$.

The elements of $\mathcal{E}_{(\omega)}(\mathbb{R}^N)$ (resp. of $\mathcal{E}_{\{\omega\}}(\mathbb{R}^N)$) are called ω -ultradifferentiable functions of Beurling type (resp. of Roumieu type).

4.2 Notation. We will write $\mathcal{E}_*$ (resp. $\mathcal{D}_*$) if a statement holds for both $\mathcal{E}_{\{\omega\}}$ and $\mathcal{E}_{(\omega)}$ (resp. $\mathcal{D}_{\{\omega\}}$ and $\mathcal{D}_{(\omega)}$).

4.3 Examples. The following functions $\omega: [0, \infty[\rightarrow [0, \infty[$ are equivalent to weight functions (see 3.2):

- (1) $\omega(t) = t^\alpha$, $0 < \alpha < 1$,
- (2) $\omega(t) = (\log(1+t))^\beta$, $\beta > 1$,
- (3) $\omega(t) = t(\log(e+t))^{-\beta}$, $\beta > 1$.

For ω as in (1) the space $\mathcal{E}_{\{\omega\}}(\Omega)$ is the classical Gevrey class of exponent $1/\alpha$.

4.4 Proposition. $\mathcal{E}_*(\Omega)$ is a locally convex algebra with continuous multiplication.

PROOF. For given $A > 0$ and $f_1, f_2 \in \mathcal{E}_*(\Omega)$, set

$$C_l = \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K} |f_l^{(\alpha)}(x)| \exp \left(-A\varphi^* \left(\frac{|\alpha|}{A} \right) \right), \quad l = 1, 2.$$

For $\alpha, \beta \in \mathbb{N}_0^N$ with $\beta \leq \alpha$ the convexity of φ^* implies $\varphi^*(\lambda x) \leq \lambda \varphi^*(x)$ for $0 \leq \lambda \leq 1$ and hence

$$\varphi^* \left(\frac{|\beta|}{A} \right) + \varphi^* \left(\frac{|\alpha - \beta|}{A} \right) \leq \varphi^* \left(\frac{|\alpha|}{A} \right).$$

By Leibniz' rule this implies as in the proof of, e.g., Komatsu [14], 2.7:

$$|(f_1 f_2)^{(\alpha)}(x)| \leq C_1 C_2 2^{|\alpha|} \exp \left(A\varphi^* \left(\frac{|\alpha|}{A} \right) \right) \leq C_1 C_2 \exp \left(|\alpha| + A\varphi^* \left(\frac{|\alpha|}{A} \right) \right).$$

Let now L be as in 1.4 and set $B = A/L$. Then it follows from 1.4 that for large $|\alpha|$

$$|\alpha| + A\varphi^* \left(\frac{|\alpha|}{A} \right) = B \left(\frac{|\alpha|}{B} + L\varphi^* \left(\frac{|\alpha|}{BL} \right) \right) \leq B \left(\varphi^* \left(\frac{|\alpha|}{B} \right) + L \right). \quad (1)$$

Sometimes a different description of $\mathcal{E}_*(\Omega)$ is useful. It follows directly from 3.3 and the proof of the previous proposition:

4.5 Lemma. Let $\Omega \subset \mathbb{R}^N$ be open, let $K_1 \subset \overset{\circ}{K}_2 \subset K_2 \subset \dots \subset \Omega$ be an exhaustion of Ω by compact sets, choose $\chi_j \in \mathcal{D}_*(K_j)$ with $0 \leq \chi_j \leq 1$ and $\chi_j|_{K_{j-1}} \equiv 1$, and define a projective spectrum by $\varphi_j: \mathcal{D}_*(K_{j+1}) \rightarrow \mathcal{D}_*(K_j)$, $f \mapsto \chi_j f$. Then

$$\mathcal{E}_*(\Omega) = \text{proj}_{j \rightarrow \infty} \mathcal{D}_*(K_j).$$

4.6 Lemma. Let K be a compact subset of an open set $\Omega \subset \mathbb{R}^N$. Then $\mathcal{D}_*(K)$ carries the topology which is induced by $\mathcal{E}_*(\Omega)$.

PROOF. Let $L \subset \Omega$ be a compact set with $K \subset \overset{\circ}{L}$, and choose $\chi \in \mathcal{D}(L)$ with $0 \leq \chi \leq 1$ and $\chi \equiv 1$ on K . Let p be a seminorm on $\mathcal{D}_*(K)$. The inclusion $\mathcal{D}_*(K) \hookrightarrow \mathcal{D}_*(L)$ is a topological homomorphism by the open mapping theorem, since $\mathcal{D}_*(K)$ is closed in $\mathcal{D}_*(L)$. Thus there is a seminorm q on $\mathcal{D}_*(L)$ with $q|_{\mathcal{D}_*(K)} = p$. Now 4.5 implies that $f \mapsto q(\chi f)$ is a seminorm on $\mathcal{E}_*(\Omega)$. Its restriction to $\mathcal{D}_*(K)$ is p .

The next proposition follows in the usual way from 3.3, 3.8, and the definitions of $\mathcal{D}_*$ and $\mathcal{E}_*$.

4.7 Proposition.

(1) The inclusion $\mathcal{D}_*(\Omega) \hookrightarrow \mathcal{E}_*(\Omega)$ is continuous and has dense range.

(2) If $\sigma = o(\omega)$, then the inclusion $\mathcal{E}_{\{\omega\}}(\Omega) \hookrightarrow \mathcal{E}_{\{\sigma\}}(\Omega)$ is continuous and has dense range.

4.8 Proposition. *Let $(\Omega_j)_{j \in \mathbb{N}}$ be an open covering of Ω . Then there are $f_j \in \mathcal{D}_*(\Omega_j)$ with $0 \leq f_j \leq 1$ such that $\sum_{j=1}^{\infty} f_j = 1$ and $(\text{Supp}(f_j))_{j \in \mathbb{N}}$ is a locally finite cover of Ω .*

PROOF. The proof of Komatsu [14], 5.2, works also in our case.

From 4.5 and 3.6 it follows easily that $\mathcal{E}_*(\Omega)$ is complete and nuclear for each open set Ω in $\mathbb{R}^N$. More information on $\mathcal{E}_*(\Omega)$ can be obtained if we apply a general result of Vogt [27]. To formulate this, we define the notion of a power series space.

For an increasing unbounded sequence α and $R = 1, \infty$, the space $\Lambda_R(\alpha)$ is defined as

$$\Lambda_R(\alpha) = \left\{ x \in \mathbb{C}^N \mid \|x\|_r := \sum_{j=1}^{\infty} |x_j| r^{\alpha_j} < \infty \text{ for all } 0 < r < R \right\}.$$

Obviously, $\Lambda_R(\alpha)$ is a Fréchet-Schwartz space.

For a locally convex space E we denote by E'_b its strong dual.

4.9 Proposition. *For each weight function ω and each open set Ω in $\mathbb{R}^N$ we have for $\alpha(\omega, N) := (\omega(j^{1/N}))_{n \in \mathbb{N}}$*

$$\mathcal{E}_{(\omega)}(\Omega) \cong \Lambda_{\infty}(\alpha(\omega, N))^{\mathbb{N}}, \quad (1)$$

$$\mathcal{E}_{\{\omega\}}(\Omega) \cong (\Lambda_1(\alpha(\omega, N))'_b)^{\mathbb{N}}. \quad (2)$$

In particular, $\mathcal{E}_{(\omega)}(\Omega)$ is a nuclear Fréchet space, while $\mathcal{E}_{\{\omega\}}(\Omega)$ is nuclear, complete, and reflexive.

PROOF. We apply Theorem 5.1 of Vogt [27] with $\mathcal{D}_*$ in the rôle of F and $\mathcal{E}_*$ in the rôle of G . Then G is a sheaf of F -modules. Most of the other hypotheses of Vogt's Theorem 5.1 are clear by definition or from our previous results. We construct the partitions of unity required in Vogt's second hypothesis first in $\mathcal{D}(\mathbb{R}^N)$, convolve them with a suitable function in $\mathcal{D}_*(\mathbb{R}^N)$ and apply 3.7. Then Vogt's theorem implies

$$\mathcal{E}_*(\Omega) \cong (\mathcal{E}_{*2\pi})^{\mathbb{N}},$$

where $\mathcal{E}_{*2\pi}$ is the space of all 2π -periodic functions in $\mathcal{E}_*(\mathbb{R}^N)$. The structure of these spaces can be calculated by estimating the Fourier coefficients, thereby using 1.1(α) and (γ) (see also Meise, Schwerdtfeger, and Taylor [20], 3.2, and Meise [19], 3.8).

Note that by 4.9 it is also possible to prove the nuclearity of $\mathcal{E}_*(\Omega)$ applying the Grothendieck-Pietsch criterion to $\Lambda_R(\omega(\alpha, N))$.

4.10 Proposition. *Let $\Omega_1, \Omega_2 \subset \mathbb{R}^N$ be given, let $g: \Omega_1 \rightarrow \Omega_2$ be real-analytic, and let $f \in \mathcal{E}_*(\Omega_2)$. Then $f \circ g \in \mathcal{E}_*(\Omega_1)$. In particular, $\mathcal{E}_*(\Omega)$ contains all real-analytic functions on Ω .*

PROOF. See Hörmander [13], proof of 8.4.1.

5 Ultradistributions

We introduce ω -ultradistributions of Beurling and of Roumieu type with compact and with arbitrary support. We show that the corresponding test functions are dense in these spaces.

5.1 Definition. Let ω be a weight function and $\Omega \subset \mathbb{R}^N$ an open set.

- (1) The elements of $\mathcal{D}_{(\omega)}(\Omega)'$ (resp. $\mathcal{D}_{\{\omega\}}(\Omega)'$) are called ω -ultradistributions of Beurling type (resp. of Roumieu type).
- (2) For an ultradistribution $T \in \mathcal{D}_*(\Omega)'$ its support $\text{Supp } T$ is the set of all points for which for all open neighborhoods U there is $\varphi \in \mathcal{D}_*(U)$ such that $\langle T, \varphi \rangle \neq 0$.

5.2 Remark. The definition of the support of an ultradistribution T does not depend on the choice of the class $\mathcal{D}_*(\Omega)$ as long as it contains T . In particular, if T is a distribution, then the support defined in 5.1 is the usual one. This follows from 3.9.

5.3 Proposition. *An ultradistribution $T \in \mathcal{D}_*(\Omega)'$ can be extended continuously to $\mathcal{E}_*(\Omega)$ if and only if $\text{Supp } T$ is a compact subset of Ω .*

PROOF. It is obvious that every ultradistribution with compact support can be extended. To prove the converse, let $T \in \mathcal{E}_*(\Omega)'$ be given. Then there are a seminorm p on $\mathcal{E}_*(\Omega)$ and a constant C with $|\langle T, f \rangle| \leq Cp(f)$ for all $f \in \mathcal{E}_*(\Omega)$. We may assume that p is given by 4.5, i.e., there are a compact $K \subset \Omega$, $\chi \in \mathcal{D}_*(K)$, and a seminorm q on $\mathcal{D}_*(K)$ with $p(f) = q(\chi f)$ for all $f \in \mathcal{E}_*(\Omega)$. Thus, if U is an open subset of $\Omega \setminus K$ and $f \in \mathcal{D}_*(U)$, then we have $|\langle T, f \rangle| \leq Cq(\chi f) = 0$, which implies $\text{Supp } T \subset K$.

5.4 Proposition. *If Ω is an open set in $\mathbb{R}^N$, then $\mathcal{D}_*(\Omega)$ is dense in $\mathcal{D}_*(\Omega)'_b$ and in $\mathcal{E}_*(\Omega)'_b$.*

PROOF. Note that $\mathcal{D}_*(\Omega)$ and $\mathcal{E}_*(\Omega)$ are semireflexive by 3.6 and 4.9. We will only give the proof for $\mathcal{D}_*$, but the arguments also work for $\mathcal{E}_*$. Let $\iota: \mathcal{D}_*(\Omega) \hookrightarrow \mathcal{D}_*(\Omega)'_b$ denote the inclusion defined by $\langle \iota(f), \varphi \rangle = \int_{\Omega} f\varphi$. To show that its range is dense, let $T \in (\mathcal{D}_*(\Omega)'_b)'$ with $T|_{\text{im } \iota} = 0$ be given. Since $\mathcal{D}_*(\Omega)$ is semireflexive, T corresponds to $f \in \mathcal{D}_*(\Omega)$ with $\int_{\Omega} f\varphi = 0$ for all $\varphi \in \mathcal{D}_*(\Omega)$. This means that f and hence T are zero.

5.5 Definition. Let $\Omega \subset \mathbb{R}^N$ be open. For $f \in \mathcal{E}_*(\Omega)$ and $T \in \mathcal{D}_*(\Omega)'$ we define $fT \in \mathcal{D}_*(\Omega)'$ by

$$\langle fT, \varphi \rangle = \langle T, f\varphi \rangle \quad \text{for all } \varphi \in \mathcal{D}_*(\Omega).$$

This makes $\mathcal{D}_*(\Omega)'$ an $\mathcal{E}_*(\Omega)$ -module.

The linear topological structure of $\mathcal{D}_*(\Omega)'_b$ is completely described by the following proposition.

5.6 Proposition. *For each weight function ω and each open set Ω in $\mathbb{R}^N$, we have for $\alpha(\omega, N) := (\omega(j^{1/N}))_{j \in \mathbb{N}}$:*

$$\mathcal{D}_{\{\omega\}}(\Omega)'_b \cong \left(\Lambda_\infty(\alpha(\omega, N))'_b \right)^N, \quad (1)$$

$$\mathcal{D}_{\{\omega\}}(\Omega)'_b \cong \Lambda_1(\alpha(\omega, N))^N. \quad (2)$$

In particular, $\mathcal{D}_{\{\omega\}}(\Omega)'_b$ is a nuclear Fréchet space, while $\mathcal{D}_{\{\omega\}}(\Omega)'_b$ is complete, nuclear, and reflexive. In view of the reflexivity of $\mathcal{D}_(\Omega)$ this implies*

$$\mathcal{D}_{\{\omega\}}(\Omega) \cong \bigoplus_{\mathbb{N}} \Lambda_\infty(\alpha(\omega, N)) \quad \text{and} \quad \mathcal{D}_{\{\omega\}}(\Omega) \cong \bigoplus_{\mathbb{N}} \Lambda_1(\alpha(\omega, N))'_b.$$

PROOF. We apply Vogt [27], Theorem 5.1 with F as in the proof of 4.9. The structure of the corresponding spaces of periodic functions can be found in Braun and Meise [5], 11., for Roumieu type, and in Franken and Meise [11] for Beurling type.

6 Convolutions

For $\mu \in \mathcal{E}_*(\mathbb{R}^N)$, convolution operators on $\mathcal{E}_*(\mathbb{R}^N)$ and on $\mathcal{D}_*(\mathbb{R}^N)'$ are defined.

6.1 Definition. For a weight function ω , an ultradistribution $\mu \in \mathcal{E}_*(\mathbb{R}^N)'$, and $f \in \mathcal{E}_*(\mathbb{R}^N)$ we define the convolution by

$$T_\mu(f) := \mu * f: \mathbb{R}^N \rightarrow \mathbb{C}, \quad \mu * f(x) = \langle \mu_y, f(x - y) \rangle.$$

For $\nu \in \mathcal{D}_*(\mathbb{R}^N)'$ we define

$$S_\mu(\nu) := \mu * \nu: \mathcal{D}_*(\mathbb{R}^N) \rightarrow \mathbb{C}, \quad \langle \mu * \nu, f \rangle = \langle \nu, \check{\mu} * f \rangle, \quad f \in \mathcal{D}_*(\mathbb{R}^N),$$

where $\langle \check{\mu}, f \rangle = \langle \mu_x, f(-x) \rangle$.

6.2 Remark. If in the above the ultradistribution ν is given by a function $f \in \mathcal{E}_*(\mathbb{R}^N)$, i.e., $\langle \nu, \varphi \rangle = \int_{\mathbb{R}^N} f(x) \varphi(x) dm_N(x)$, then $T_\mu(f) = S_\mu(f)$, i.e., both definitions of $\mu * f$ coincide.

PROOF. Let $\varphi \in \mathcal{D}_*(\mathbb{R}^N)$ be arbitrary. Then

$$\begin{aligned} \langle \mu * \nu, \varphi \rangle &= \langle \nu, \check{\mu} * \varphi \rangle = \int_{\mathbb{R}^N} f(x) \check{\mu} * \varphi(x) dm_N(x) \\ &= \int_{\mathbb{R}^N} f(x) \langle \mu_y, \varphi(x + y) \rangle dm_N(x) \\ &= \int_{\mathbb{R}^N} \langle \mu_y, f(t - y) \varphi(t) \rangle dm_N(t) = \langle \mu * f, \varphi \rangle. \end{aligned}$$

6.3 Proposition. *The maps*

$$T_\mu: \mathcal{E}_*(\mathbb{R}^N) \rightarrow \mathcal{E}_*(\mathbb{R}^N)$$

and

$$S_\mu: \mathcal{D}_*(\mathbb{R}^N)'_b \rightarrow \mathcal{D}_*(\mathbb{R}^N)'_b$$

are continuous.

PROOF. First we consider T_μ . To do so, we define for $\lambda > 0$ and an open convex bounded set $K \subset \mathbb{R}^N$ the Banach space

$$\mathcal{E}_\lambda(K) = \left\{ f \in C^\infty(K) \mid \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K} |f^{(\alpha)}(x)| \exp \left(-\lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) < \infty \right\}.$$

If we let $K_j = \{x \in \mathbb{R}^N \mid |x| < j\}$, then

$$\mathcal{E}_{\{\omega\}}(\mathbb{R}^N) = \text{proj}_{j \rightarrow \infty} \text{ind}_{\lambda \rightarrow 0} \mathcal{E}_\lambda(K_j) \quad \text{and} \quad \mathcal{E}_{\{\omega\}}(\mathbb{R}^N) = \text{proj}_{j \rightarrow \infty} \text{proj}_{\lambda \rightarrow \infty} \mathcal{E}_\lambda(K_j).$$

If $\mu \in \mathcal{E}_{\{\omega\}}(\mathbb{R}^N)'$ then there is $k \in \mathbb{N}$ such that for all $\lambda > 0$ we have $\mu \in \mathcal{E}_\lambda(K_k)'$. If $\mu \in \mathcal{E}_{\{\omega\}}(\mathbb{R}^N)'$, then there are $k \in \mathbb{N}$ and $\lambda_0 > 0$ such that $\mu \in \mathcal{E}_\lambda(K_k)'$ for all $\lambda \geq \lambda_0$. In both cases we show the continuity of

$$T_\mu: \mathcal{E}_{2\lambda}(K_{j+k}) \rightarrow \mathcal{E}_\lambda(K_j)$$

for all j and all $\lambda > 0$ in case $\mu \in \mathcal{E}_{\{\omega\}}(\mathbb{R}^N)'$ and all j and all $\lambda \geq \lambda_0$ in case $\mu \in \mathcal{E}_{\{\omega\}}(\mathbb{R}^N)'$. First note that for $\alpha, \beta \in \mathbb{N}_0^N$ and $\lambda > 0$ we have

$$\varphi^* \left(\frac{|\alpha + \beta|}{2\lambda} \right) = \varphi^* \left(\frac{1}{2} \frac{|\alpha|}{\lambda} + \frac{1}{2} \frac{|\beta|}{\lambda} \right) \leq \frac{1}{2} \varphi^* \left(\frac{|\alpha|}{\lambda} \right) + \frac{1}{2} \varphi^* \left(\frac{|\beta|}{\lambda} \right). \quad (1)$$

Using this and (1) out of the proof of 4.4 it is easy to see that for $f \in \mathcal{E}_*(\mathbb{R}^N)$ and $e = (1, 0, \dots, 0)$ the functions $\frac{1}{h}(f(\cdot + he) - f)$ converge to $\partial f / \partial x_1$ in $\mathcal{E}_*(\mathbb{R}^N)$. This implies

$$\begin{aligned} |(\mu * f)^{(\alpha)}(x)| &= |\langle \mu_y, f^{(\alpha)}(x - y) \rangle| \\ &\leq \sup_{\beta \in \mathbb{N}_0^N} \sup_{|y| < k} |f^{(\alpha + \beta)}(x - y)| \exp \left(-\lambda \varphi^* \left(\frac{|\beta|}{\lambda} \right) \right). \end{aligned}$$

Because of (1), this implies

$$|(\mu * f)^{(\alpha)}(x)| \leq \sup_{\beta \in \mathbb{N}_0^N} \sup_{|y| < k} |f^{(\alpha + \beta)}(x - y)| \exp \left(-2\lambda \varphi^* \left(\frac{|\alpha + \beta|}{2\lambda} \right) + \lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right).$$

Hence we have

$$\begin{aligned} &\sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K_j} |(\mu * f)^{(\alpha)}(x)| \exp \left(-\lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) \\ &\leq \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K_j} \sup_{\beta \in \mathbb{N}_0^N} \sup_{|y| < k} |f^{(\alpha + \beta)}(x - y)| \exp \left(-2\lambda \varphi^* \left(\frac{|\alpha + \beta|}{2\lambda} \right) \right) \\ &\leq \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K_{j+k}} |f^{(\alpha)}(x)| \exp \left(-2\lambda \varphi^* \left(\frac{|\alpha|}{2\lambda} \right) \right). \end{aligned}$$

For the proof of the continuity of S_μ we consider its adjoint $S_\mu^t: \mathcal{D}_*(\mathbb{R}^N) \rightarrow \mathcal{D}_*(\mathbb{R}^N)$ and proceed as above, but replace $\mathcal{E}_\lambda(K)$ by

$$\mathcal{D}_\lambda(K) = \left\{ f \in \mathcal{D}(K) \mid \sup_{\alpha \in \mathbb{N}_0^N} \sup_{x \in K} |f^{(\alpha)}(x)| \exp \left(-\lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) < \infty \right\}$$

thus applying the description of $\mathcal{D}_*(\mathbb{R}^N)$ given in the Paley-Wiener theorem 3.4.

6.4 Proposition. $\mathcal{D}_*(\mathbb{R}^N) * \mathcal{D}_*(\mathbb{R}^N)'$ is a subset of $\mathcal{E}_*(\mathbb{R}^N)$. For $f \in \mathcal{D}_*(\mathbb{R}^N)$ and $\nu \in \mathcal{D}_*(\mathbb{R}^N)'$ we have $f * \nu(x) = \langle \nu_y, f(x - y) \rangle$.

PROOF. We show first that for $f, \varphi \in \mathcal{D}_*(\mathbb{R}^N)$ the map

$$\mathcal{D}_*(\mathbb{R}^N)'_b \rightarrow \mathbb{C}, \quad \nu \mapsto \langle \langle \nu_y, f(x - y) \rangle, \varphi_x \rangle,$$

is continuous. It is easy to see that the set $B = \{y \mapsto f(x - y) \mid x \in \text{Supp } \varphi\}$ is bounded in $\mathcal{D}_*(\mathbb{R}^N)$. Hence $p(\nu) := \sup_{f \in B} |\langle \nu, f \rangle|$ is a continuous seminorm on $\mathcal{D}_*(\mathbb{R}^N)'_b$. Thus

$$|\langle \langle \nu_y, f(x - y) \rangle, \varphi_x \rangle| \leq \int_{\text{Supp } \varphi} |\langle \nu_y, f(x - y) \rangle \varphi(x)| dm_N(x) \leq p(\nu) \int |\varphi|.$$

Let now $\nu \in \mathcal{D}_*(\mathbb{R}^N)'$ and $f, \varphi \in \mathcal{D}_*(\mathbb{R}^N)$ be given. Choose a net $(g_\alpha)_\alpha$ in $\mathcal{D}_*(\mathbb{R}^N)$ that converges in $\mathcal{D}_*(\mathbb{R}^N)'_b$ to ν . Then

$$\begin{aligned} \langle f * \nu, \varphi \rangle &= \langle \nu, \check{f} * \varphi \rangle = \lim_{\alpha \rightarrow \infty} \langle g_\alpha, \check{f} * \varphi \rangle = \lim_{\alpha \rightarrow \infty} \int_{\mathbb{R}^{2N}} g_\alpha(x) f(y) \varphi(x + y) dm_{2N}(x, y) \\ &= \lim_{\alpha \rightarrow \infty} \langle \langle (g_\alpha)_y, f(x - y) \rangle, \varphi_x \rangle = \langle \langle \nu_y, f(x - y) \rangle, \varphi_x \rangle. \end{aligned}$$

Thus the ultradistribution $f * \nu$ is given by the function $x \mapsto \langle \nu_y, f(x - y) \rangle$. To show that it is ultradifferentiable, let $x \in \mathbb{R}^N$ be given, and choose $\psi \in \mathcal{D}_*(\mathbb{R}^N)$ with $\psi(y) = 1$ for all y with $|y - (x - a)| < 1$ for some $a \in \text{Supp } f$. Then we have for $|\xi - x| < 1$:

$$\langle \nu_y, f(\xi - y) \rangle = \langle \nu_y, \psi(y) f(\xi - y) \rangle = \langle \psi \nu, f(\xi - y) \rangle = (\psi \nu) * f(\xi).$$

Since $\psi \nu \in \mathcal{E}_*(\mathbb{R}^N)'$, its convolution with f is ultradifferentiable.

7 The Fourier-Laplace transform

For open and convex $\Omega \subset \mathbb{R}^N$ we identify $\mathcal{E}_*(\Omega)'_b$ via Fourier-Laplace transform with a space of entire functions depending on ω and Ω . Furthermore, we show that for every ω -ultradistribution μ of Roumieu type there is a weight function σ with $\mathcal{D}_{(\sigma)}(\Omega) \subset \mathcal{D}_{\{\omega\}}(\Omega)$ such that μ is a σ -ultradistribution of Beurling type.

7.1 Definition. For $z \in \mathbb{C}^N$ we let

$$f_z(x) = \exp(-i\langle x, z \rangle), \quad x \in \mathbb{R}^N.$$

For each $\lambda > 0$ we have

$$\begin{aligned} \sup_{\alpha \in \mathbb{N}_0^N} |f_z^{(\alpha)}(x)| \exp \left(-\lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) &= \sup_{\alpha \in \mathbb{N}_0^N} |z^\alpha| |\exp(-i\langle x, z \rangle)| \exp \left(-\lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) \\ &\leq \exp \left(\langle x, \operatorname{Im} z \rangle + \sup_{\alpha \in \mathbb{N}_0^N} |\alpha| \log |z| - \lambda \varphi^* \left(\frac{|\alpha|}{\lambda} \right) \right) \leq \exp(\langle x, \operatorname{Im} z \rangle + \lambda \omega(z)). \end{aligned}$$

Thus $f_z \in \mathcal{E}_*(\Omega)$ for all ω and Ω . The Fourier-Laplace transform $\hat{\mu}$ of $\mu \in \mathcal{E}_*(\Omega)'$ is given by

$$\hat{\mu}: z \mapsto \langle \mu, f_z \rangle.$$

Note that for $\varphi \in \mathcal{D}_*(\Omega)$:

$$\begin{aligned} \widehat{\mu * \varphi}(z) &= \int_{\mathbb{R}^N} \mu * \varphi(t) f_z(t) dm_N(t) = \int_{\mathbb{R}^N} \langle \mu_y, f_z(t) \varphi(t - y) \rangle dm_N(t) \\ &= \int_{\mathbb{R}^N} \langle \mu_y, f_z(s + y) \varphi(s) \rangle dm_N(s) = \int_{\mathbb{R}^N} \langle \mu, f_z \rangle \varphi(s) f_z(s) dm_N(s) = \hat{\mu}(z) \hat{\varphi}(z). \end{aligned}$$

7.2 Proposition. *Let $\Omega \subset \mathbb{R}^N$ be open and suppose that for $\mu \in \mathcal{E}_{(\omega)}(\Omega)'$ there are $\lambda, C > 0$ and a compact $K \subset \Omega$ such that*

$$|\langle \mu, f \rangle| \leq C p_{K, \lambda}(f) \quad \text{for all } f \in \mathcal{E}_{(\omega)}(\mathbb{R}^N). \quad (1)$$

Then $\hat{\mu}$ is entire and satisfies

$$|\hat{\mu}(z)| \leq C \exp(H_K(\operatorname{Im} z) + \lambda \omega(z)) \quad \text{for all } z \in \mathbb{C}^N, \quad (2)$$

$$\langle \mu, \varphi \rangle = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \hat{\mu}(-t) \hat{\varphi}(t) dm_N(t) \quad \text{for all } \varphi \in \mathcal{D}_{(\omega)}(\mathbb{R}^N). \quad (3)$$

PROOF. We had seen in 7.1 that for every compact $L \subset \mathbb{R}^N$

$$p_{L, \lambda}(f_z) \leq \exp(H_L(\operatorname{Im} z) + \lambda \omega(z)) \quad \text{for } z \in \mathbb{C}^N, \lambda > 0. \quad (4)$$

This proves (2). We define $F(z) = f_z$. We claim that F is holomorphic. In view of the identity $F(z + w) = F(z)F(w)$ for all $z, w \in \mathbb{C}^N$ it is enough to prove differentiability in the origin. For $x \in \mathbb{R}^N$, $z \in \mathbb{C}^N$ with $|z| < 1$, and $k \in \mathbb{N}$ we have

$$\begin{aligned} |D^\alpha(f_z(x) - 1 + i\langle x, z \rangle)| &= \left| \sum_{k=\max\{2, |\alpha|\}}^{\infty} \frac{(-i\langle x, z \rangle)^{k-|\alpha|}}{(k-\alpha)!} (-iz)^\alpha \right| \\ &\leq \sum_{k=\max\{2, |\alpha|\}}^{\infty} \frac{|x|^{k-|\alpha|}}{(k-|\alpha|)!} |z|^k \leq |z|^2 e^{|x|}. \end{aligned}$$

Thus $\hat{\mu} = \mu \circ F$ is entire. For $\varphi \in \mathcal{D}_{(\omega)}(\Omega)$ set $\check{\varphi}(x) = \varphi(-x)$. Then $\hat{\check{\varphi}}(t) = \varphi(-t)$, and (3) is shown in the following way

$$\begin{aligned} \langle \mu, \varphi \rangle &= \mu * \check{\varphi}(0) = \mathcal{F}^{-1}(\widehat{\mu * \check{\varphi}})(0) = \mathcal{F}^{-1}(\hat{\mu} \hat{\check{\varphi}})(0) \\ &= \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \hat{\mu}(t) \hat{\varphi}(-t) dm_N(t) = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} \hat{\mu}(-t) \hat{\varphi}(t) dm_N(t). \end{aligned}$$

7.3 Proposition. *Suppose that for $g \in A(\mathbb{C}^N)$ there are a compact convex $K \subset \mathbb{R}^N$, $C > 0$, and $m \in \mathbb{N}$ such that*

$$|g(z)| \leq C \exp(H_K(\operatorname{Im} z) + m\omega(z)) \quad \text{for all } z \in \mathbb{C}^N.$$

Then there is $\mu \in \mathcal{E}_{(\omega)}(\mathbb{R}^N)'$ with $\hat{\mu} = g$ and $\operatorname{Supp} \mu \subset K$.

PROOF. We define μ by

$$\langle \mu, f \rangle = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} g(-t) \hat{f}(t) dm_N(t) \quad \text{for } f \in \mathcal{D}_{(\omega)}(\mathbb{R}^N).$$

It is easy to see that $\mu \in \mathcal{D}_{(\omega)}(\mathbb{R}^N)'$. As in the classical case, $\operatorname{Supp} \mu \subset K$. Hence, by 5.3, we have $\mu \in \mathcal{E}_{(\omega)}(\mathbb{R}^N)'$. Equation (3) of 7.2 shows that $\hat{\mu} = g$.

Let a weight function ω and a convex open set $\Omega \subset \mathbb{R}^N$ be given. Choose a convex compact exhaustion $K_1 \subset \overset{\circ}{K}_2 \subset K_2 \subset \cdots \subset \Omega$. Then we define the following two spaces of entire functions.

$$A_{(\omega), \Omega}(\mathbb{C}^N) = A_{\mathbb{P}}(\mathbb{C}^N), \quad \text{where } \mathbb{P} = \{p_n: z \mapsto H_{K_n}(\operatorname{Im} z) + n\omega(z) \mid n \in \mathbb{N}\},$$

$$A_{\{\omega\}, \Omega}(\mathbb{C}^N) = \left\{ f \in H(\mathbb{C}^N) \mid \begin{array}{l} \text{there is } n \in \mathbb{N} \text{ such that for all } \varepsilon > 0 \\ \sup_{z \in \mathbb{C}^N} |f(z)| \exp(-H_{K_n}(\operatorname{Im} z) - \varepsilon \omega(z)) < \infty \end{array} \right\}.$$

The space $A_{\{\omega\}, \Omega}(\mathbb{C}^N)$ carries the natural (LF)-space topology. Both definitions do not depend on the special choice of the compact exhaustion for Ω .

7.4 Theorem. *Let $\Omega \subset \mathbb{R}^N$ be convex and open. The Fourier-Laplace transform*

$$\mathcal{F}(\mu)[z] = \hat{\mu}(z) = \langle \mu_x, \exp(-i\langle x, z \rangle) \rangle, \quad \mu \in \mathcal{E}_*(\Omega)',$$

is a linear topological isomorphism from $\mathcal{E}_{(\omega)}(\Omega)'_b$ onto $A_{(\omega), \Omega}(\mathbb{C}^N)$ as well as from $\mathcal{E}_{\{\omega\}}(\Omega)'_b$ onto $A_{\{\omega\}, \Omega}(\mathbb{C}^N)$.

PROOF. For ultradistributions of Beurling type, the claim follows from 7.2, 7.3, and the closed graph theorem.

To prove the Roumieu case, choose for each $n \in \mathbb{N}$ a function $\chi_n \in \mathcal{D}_{\{\omega\}}(\mathbb{R}^N)$ with $0 \leq \chi_n \leq 1$, $\chi_n \equiv 1$ on K_n , and $\operatorname{Supp} \chi_n \subset \overset{\circ}{K}_{n+1}$. If we let $\mathbb{P}_n = (p_{j,n})_{j \in \mathbb{N}}$ where $p_{j,n}(z) = H_{K_n}(\operatorname{Im} z) + \omega(z)/j$ as in Chapter 3, we can define

$$G_n: A_{\mathbb{P}_n}(\mathbb{C}^N) \rightarrow \mathcal{E}_{\{\omega\}}(\Omega)'_b, \quad \langle G_n(g), f \rangle = \frac{1}{(2\pi)^N} \int_{\mathbb{R}^N} g(-t) \widehat{\chi_n f}(t) dm_N(t) \quad \text{for } f \in \mathcal{E}_{\{\omega\}}(\Omega).$$

If $g \in A_{\mathbb{P}_n}(\mathbb{C}^N)$, then $\operatorname{Supp} G_{n+1}(g) \subset K_n$, hence

$$\langle G_{n+1}(g), f \rangle = \langle G_{n+1}(g), \chi_n f \rangle = \langle G_n(g), \chi_{n+1} f \rangle = \langle G_n(g), f \rangle,$$

which implies that the restriction of G_{n+1} to $A_{\mathbb{P}_n}$ is G_n . These considerations show that once the continuity of all G_n is established, we get a continuous map $G: A_{\{\omega\}, \Omega}(\mathbb{C}^N) \rightarrow \mathcal{E}_{\{\omega\}}(\Omega)'_b$ which coincides with G_n on $A_{\mathbb{P}_n}(\mathbb{C}^N)$. To see that G_n is continuous, let a bounded set B in $\mathcal{E}_{\{\omega\}}(\Omega)$ be given. By 4.5 we have $\mathcal{E}_{\{\omega\}}(\Omega) = \text{proj}_{j \rightarrow \infty} \mathcal{D}_{\{\omega\}}(K_j)$. By 3.5 we can identify $\mathcal{D}_{\{\omega\}}(K_j)$ with a (DFN)-space $A_{\mathbb{P}}(\mathbb{C}^N)$. The statement that B is bounded then reads as follows

$$\forall j \exists m, C \forall f \in B : \sup_{z \in \mathbb{C}^N} |\widehat{\chi_j f}(z)| \exp \left(-H_{j+1}(\text{Im } z) + \frac{1}{m} \omega(z) \right) \leq C. \quad (1)$$

We let $j = n$ and choose m and C according to (1). For $g \in A_{\mathbb{P}_n}$ with

$$\sup_{z \in \mathbb{C}^N} |g(z)| \exp \left(-H_n(\text{Im } z) - \frac{1}{2m} \omega(z) \right) \leq 1$$

we have

$$\begin{aligned} \sup_{f \in B} |\langle G_n(g), f \rangle| &= \sup_{f \in B} \frac{1}{(2\pi)^N} \left| \int_{\mathbb{R}^N} g(-t) \widehat{\chi_n f}(t) dm_N(t) \right| \\ &\leq \frac{C}{(2\pi)^N} \int_{\mathbb{R}^N} \exp \left(\frac{1}{2m} \omega(t) - \frac{1}{m} \omega(t) \right) dm_N(t). \end{aligned}$$

Thus G_n is continuous. Since $\mathcal{D}_{\{\omega\}}(\Omega)$ is dense in $\mathcal{D}_{\{\omega\}}(\Omega)$, 7.2(3) shows that $G_n \circ \mathcal{F} = \text{id}$, and hence $G = \mathcal{F}^{-1}$. We had seen in 4.9 that $\mathcal{E}_{\{\omega\}}(\Omega)$ is a complete Schwartz space, hence its dual is ultrabornological by Schwartz [26], p. 43. Thus, DeWilde's open mapping theorem (see, e.g., Köthe [16], §35.3.(5)) implies the continuity of $\mathcal{F}$.

7.5 Proposition. *Let $\mu, \nu \in \mathcal{E}_*(\mathbb{R}^N)'$ be given and denote by $M_{\hat{\mu}}^\wedge$ the multiplication operator $M_{\hat{\mu}}^\wedge(f) = f\hat{\mu}$. Then*

$$(1) \quad T_\mu^t = \mathcal{F}^{-1} \circ M_{\hat{\mu}}^\wedge \circ \mathcal{F},$$

$$(2) \quad (\ker T_\mu)_b' \cong A_* / \overline{\hat{\mu} A_*}.$$

In the Beurling case, (2) implies

$$(3) \quad \ker T_\mu \cong (A_{(\omega)} / \overline{\hat{\mu} A_{(\omega)}})_b'.$$

PROOF. (1) is verified by direct calculation. In the Beurling case, (2) and (3) follow from duality theory, while in the Roumieu case (2) is proved in Meise [19], 3.4.

7.6 Proposition. *Let $\Omega \subset \mathbb{R}^N$ be open. For every $\mu \in \mathcal{D}_{\{\omega\}}(\Omega)'$ there is a weight function σ with $\sigma = o(\omega)$ such that $\mu \in \mathcal{D}_{(\sigma)}(\Omega)' \subset \mathcal{D}_{\{\omega\}}(\Omega)'$. The analogous statement holds for $\mathcal{E}_{\{\omega\}}(\Omega)'$.*

PROOF. Let $K_1 \subset \overset{\circ}{K}_2 \subset K_2 \subset \dots$ be an exhaustion of Ω by compact sets. A given $\mu \in \mathcal{D}_{\{\omega\}}(\Omega)'$ can be written in the form

$$\mu = \sum_{j=1}^{\infty} \mu_j \quad \text{with } \mu_j \in \mathcal{E}_{\{\omega\}}(\Omega)', \text{ Supp } \mu_j \subset K_{j+2} \setminus K_j.$$

Define for each $j \in \mathbb{N}$

$$g_j(r) = \sup_{|z|=r} \log |\hat{\mu}_j(z)| - H_{K_j}(\operatorname{Im} z) \quad \text{for } r > 0, \quad (1)$$

then $g_j = o(\omega)$ for all j because of 7.4. Hence by Proposition 1.8 there is a weight function σ with $\sigma = o(\omega)$ and $g_j = o(\sigma)$ for all j . If we denote by Γ the convex hull of Ω then Theorem 7.4 and (1) imply that $\mu_j \in \mathcal{E}_{(\sigma)}(\Gamma)'$. On the other hand, $\operatorname{Supp} \mu_j \subset K_j$, so that the sum $\sum_{j=1}^{\infty} \mu_j$ is locally finite in Ω and, hence, convergent in $D_{(\sigma)}(\Omega)'_b$. The second part of the statement follows from the first one and 5.2.

8 Concluding remarks

In this section we indicate how the kernel theorem of L. Schwartz extends to the present classes of ultradifferentiable functions and ultradistributions. Moreover, we use Ciorănescu and Zsidó [9], section 7, to compare the present theory with existing other ones.

Using well known results of Grothendieck [12] on topological tensor products one can prove the following theorem (see, e.g., Komatsu [15], 2.1 and 2.3, and Meyer [23]).

8.1 Theorem. *Let ω be a weight function, let $K_j \subset \mathbb{R}^{N_j}$ be compact and $\Omega_j \subset \mathbb{R}^{N_j}$ be open for $j = 1, 2$. Then we have the following canonical isomorphisms*

$$\begin{aligned} \mathcal{D}_*(K_1) \hat{\otimes}_{\pi} \mathcal{D}_*(K_2) &\cong \mathcal{D}_*(K_1 \times K_2) \\ \mathcal{D}_{\{\omega\}}(\Omega_1) \hat{\otimes}_{\pi} \mathcal{D}_{\{\omega\}}(\Omega_2) &\cong \mathcal{D}_{\{\omega\}}(\Omega_1 \times \Omega_2) \\ \mathcal{E}_*(\Omega_1) \hat{\otimes}_{\pi} \mathcal{E}_*(\Omega_2) &\cong \mathcal{E}_*(\Omega_1 \times \Omega_2) \\ \mathcal{D}_*(\Omega_1)'_b \hat{\otimes}_{\pi} \mathcal{D}_*(\Omega_2)'_b &\cong \mathcal{D}_*(\Omega_1 \times \Omega_2)'_b \end{aligned}$$

To compare different approaches to ultradifferentiable functions, we recall from Ciorănescu and Zsidó [9], section 7:

8.2 Theories of ultradistributions. Let Σ be an index set and assume that for each $\sigma \in \Sigma$ a locally convex space $\mathcal{D}_{\sigma} \subset \mathcal{D}(\mathbb{R})$ is defined, so that the following conditions are satisfied.

- (1) $\mathcal{D}_{\sigma}$ is an (LF)-space,
- (2) the topology of $\mathcal{D}_{\sigma}$ is stronger than the topology of pointwise convergence,
- (3) $\mathcal{D}_{\sigma}$ is an algebra under pointwise multiplication,
- (4) for every compact $K \subset \mathbb{R}$ and open $D \supset K$ there exists $\varphi \in \mathcal{D}_{\sigma}$ such that $0 \leq \varphi \leq 1$, $\varphi(x) = 1$ for all $x \in K$, $\operatorname{Supp} \varphi \subset D$,

- (5) the space $A(\mathbb{R})$ of all complex valued real analytic functions on $\mathbb{R}$ is dense in the space $\mathcal{E}_\sigma$, which is defined as

$$\mathcal{E}_\sigma = \left\{ f \in C^\infty(\mathbb{R}) \mid fg \in \mathcal{D}_\sigma \text{ for each } g \in \mathcal{D}_\sigma \right\},$$

endowed with the projective topology with respect to the maps $(M_g)_{g \in \mathcal{D}_\sigma}$, $M_g: \mathcal{E}_\sigma \rightarrow \mathcal{D}_\sigma$, $M_g(f) := fg$.

Then $\{\mathcal{D}_\sigma\}_{\sigma \in \Sigma}$ is called a theory of ultradistributions and the elements of $\mathcal{D}'_\sigma$ are called σ -ultradistributions.

If $\{\mathcal{D}_\sigma\}_{\sigma \in \Sigma}$ and $\{\mathcal{D}_\tau\}_{\tau \in T}$ are theories of ultradistributions, then $\{\mathcal{D}_\tau\}_{\tau \in T}$ is called larger than $\{\mathcal{D}_\sigma\}_{\sigma \in \Sigma}$ if for each $\sigma \in \Sigma$ there exists $\tau \in T$ so that $\mathcal{D}_\tau \subset \mathcal{D}_\sigma$, which is equivalent to $\mathcal{E}_\tau \subset \mathcal{E}_\sigma$ (see [9], 7.2). The two theories are equivalent if each one is larger than the other.

8.3 Proposition. Denote by $\mathcal{W}$ the set of all weight functions and let $\mathcal{D}_{(\omega)} := \mathcal{D}_{(\omega)}(\mathbb{R})$ and $\mathcal{D}_{\{\omega\}} := \mathcal{D}_{\{\omega\}}(\mathbb{R})$ for $\omega \in \mathcal{W}$. Then $(\mathcal{D}_{(\omega)})_{\omega \in \mathcal{W}}$ and $(\mathcal{D}_{\{\omega\}})_{\omega \in \mathcal{W}}$ are equivalent theories of ultradistributions.

PROOF. Conditions 8.2(1) and 8.2(2) are obvious consequences of 3.4 and 5.6. Condition 8.2(3) holds by 4.4, while 8.2(4) follows from 3.7. Condition 8.2(5) is a consequence of 3.4 and 7.4. The equivalence of the two theories follows from 3.9 and 1.6.

8.4 Other theories of ultradistributions. By Ciorănescu and Zsidó [9], 7.3, the following examples are theories of ultradistributions.

- (1) Denjoy-Carleman theory (see Roumieu [25], Lions-Magenes [17], Chou [8], Komatsu [14]): Let $\mathcal{M}$ denote the set of all positive sequences $(M_p)_{p \in \mathbb{N}_0}$ with $M_0 = 1$ which satisfy

$$(M1) \quad M_p^2 \leq M_{p-1} M_{p+1} \text{ for all } p \in \mathbb{N},$$

$$(M3)' \quad \sum_{p=1}^{\infty} \frac{M_{p-1}}{M_p} < \infty.$$

For $(M_p)_{p \in \mathbb{N}_0}$ in $\mathcal{M}$ define

$$\mathcal{D}_{\{M_p\}} = \left\{ f \in \mathcal{D}(\mathbb{R}) \mid \text{there exists } h > 0: \sup_{p \in \mathbb{N}_0} \sup_{x \in \mathbb{R}} \frac{|f^{(p)}(x)|}{h^p M_p} < \infty \right\},$$

$$\mathcal{D}_{(M_p)} = \left\{ f \in \mathcal{D}(\mathbb{R}) \mid \text{for each } h > 0: \sup_{p \in \mathbb{N}_0} \sup_{x \in \mathbb{R}} \frac{|f^{(p)}(x)|}{h^p M_p} < \infty \right\}.$$

- (2) Beurling-Björck theory (see Beurling [1], Björck [2]): Let $\mathcal{A}$ denote the set of all functions $\alpha: [0, \infty[\rightarrow [0, \infty[$ which have the following properties

$$\begin{aligned} \alpha(0) &= 0 & \alpha(t+s) &\leq \alpha(t) + \alpha(s) \quad \forall t, s > 0 \\ \int_{-\infty}^{+\infty} \frac{\alpha(t)}{1+t^2} dt &< \infty & \log t &= O(\alpha(t)) \end{aligned}$$

For $\alpha \in \mathcal{A}$ define

$$\mathcal{D}_\alpha = \left\{ f \in \mathcal{D}(\mathbb{R}) \mid \int_{-\infty}^{+\infty} |\hat{f}(t)| e^{\lambda \alpha(t)} dt < \infty \text{ for all } \lambda > 0 \right\}.$$

- (3) Ciorănescu-Zsidó theory (see [9]): Let $(t_k)_{k \in \mathbb{N}}$ be a sequence of positive numbers with $\sum_{k=1}^{\infty} 1/t_k < \infty$. Then

$$\omega_{\{t_k\}}: z \mapsto \prod_{k=1}^{\infty} \left(1 + \frac{iz}{t_k}\right)$$

is an entire function of exponential type zero. Denote by Ω the set of all these functions and let

$$\Omega_0 = \left\{ \omega_{\{t_k\}} \mid (t_k)_{k \in \mathbb{N}} \text{ satisfies } \frac{t_j}{j} \leq \frac{t_{j+1}}{j+1} \text{ for all } j \in \mathbb{N} \right\}.$$

For $\omega \in \Omega$ define

$$\mathcal{D}_\omega = \left\{ f \in \mathcal{D}(\mathbb{R}) \mid \int_{-\infty}^{+\infty} |\hat{f}(t) \omega(Lt)^n| dt < \infty \text{ for all } L > 0 \text{ and all } n \in \mathbb{N} \right\}.$$

To discuss the relations between the theories which were briefly introduced in 8.4 and the one presented so far, we use results of Ciorănescu and Zsidó [9] and the following Lemma, which was communicated to the authors by U. Franken.

8.5 Lemma. *Let $(t_k)_{k \in \mathbb{N}}$ be a sequence of positive numbers with $\sum_{k=1}^{\infty} 1/t_k < \infty$. Then*

$$\omega: t \mapsto \log |\omega_{\{t_k\}}(t)|, \quad t \geq 0$$

has all properties of a weight function except that $\omega|_{[0,1]} \not\equiv 0$. In particular, ω is equivalent to a weight function.

PROOF. Obviously, ω is continuous and increasing. By Ciorănescu and Zsidó [9], Lemma 1.6 (iii) $\Rightarrow$ (i), ω satisfies condition 1.1(β). To see that also condition 1.1(α) holds, fix $L \geq 1$ and $t > 0$. Then we have

$$\begin{aligned} \omega(Lt) &= \log \left| \prod_{k=1}^{\infty} \left(1 + \frac{iLt}{t_k}\right) \right| = \sum_{k=1}^{\infty} \log \left(1 + \left(\frac{Lt}{t_k}\right)^2\right)^{1/2} \\ &= \frac{1}{2} \sum_{k=1}^{\infty} \log \left(1 + \left(\frac{Lt}{t_k}\right)^2\right) \leq \frac{L^2}{2} \sum_{k=1}^{\infty} \log \left(1 + \left(\frac{t}{t_k}\right)^2\right) \\ &= L^2 \omega(t). \end{aligned}$$

In particular, ω satisfies 1.1(α). Next note that ω and $\varphi: x \mapsto \omega(e^x)$ are differentiable and that we have for $t > 0$

$$t\omega'(t) = \frac{t}{2} \left(\sum_{k=1}^{\infty} \log \left(1 + \left(\frac{t}{t_k}\right)^2\right) \right)' = \sum_{k=1}^{\infty} \frac{t^2}{t^2 + t_k^2}.$$

Thus $\varphi'(x) = \omega'(e^x)e^x$ is increasing and unbounded. This shows that ω satisfies 1.1(γ) and 1.1(δ).

As an immediate consequence of Lemma 8.5 we get:

8.6 Corollary. *For each $\omega \in \Omega$ there exists a weight function $\tilde{\omega} \in \mathcal{W}$ so that for each open set G in $\mathbb{R}$ the space $\mathcal{D}_\omega(G)$ in the sense of Ciorănescu and Zsidó [9] coincides with $\mathcal{D}_{(\tilde{\omega})}(G)$ defined in 3.1.*

Remark. Corollary 8.6 shows that the theory of ultradistributions presented here extends the one of Ciorănescu and Zsidó to $\mathbb{R}^N$. Hence problem 7.9(v) of [9] is solved as far as $\mathbb{R}^N$ is concerned.

From Ciorănescu and Zsidó [9], 1.6, Lemma 8.5, and Corollary 2.6 we get:

8.7 Corollary. *For each increasing continuous function $\sigma: [0, \infty[\rightarrow [0, \infty[$ which satisfies $\int_1^\infty \sigma(t)/t^2 dt < \infty$ there exist a weight function ω and $A > 0$ so that*

$$\sigma(t) \leq \omega(t) + A \quad \text{for all } t \geq 0.$$

In particular, there exists $f \in \mathcal{D}(\mathbb{R})$, $f \neq 0$, such that

$$\int_{-\infty}^{+\infty} |\hat{f}(t)| \exp(m\sigma(t)) dt < \infty \quad \text{for all } m \in \mathbb{N}.$$

By 1.6, 7.4, and 7.7 of Ciorănescu and Zsidó [9] and Corollary 8.6 we have the following relations among the theories of ultradistributions mentioned so far:

8.8 Proposition.

- (1) *The theories $\{\mathcal{D}_\omega\}_{\omega \in \Omega}$, $\{\mathcal{D}_{(M_p)}\}_{(M_p) \in \mathcal{M}}$, $\{\mathcal{D}_{\{M_p\}}\}_{(M_p) \in \mathcal{M}}$, $\{\mathcal{D}_{(\omega)}\}_{\omega \in \mathcal{W}}$, and $\{\mathcal{D}_{\{\omega\}}\}_{\omega \in \mathcal{W}}$ are equivalent.*
- (2) *The theories $\{\mathcal{D}_\omega\}_{\omega \in \Omega_0}$ and $\{\mathcal{D}_\alpha\}_{\alpha \in \mathcal{A}}$ are equivalent. $\{\mathcal{D}_{(\omega)}\}_{\omega \in \mathcal{W}}$ and $\{\mathcal{D}_{\{\omega\}}\}_{\omega \in \mathcal{W}}$ are strictly larger than these.*

Remark. The fact that $\{\mathcal{D}_{(\omega)}\}_{\omega \in \mathcal{W}}$ is strictly larger than $\{\mathcal{D}_\alpha\}_{\alpha \in \mathcal{A}}$ is also a consequence of Franken [10], S 7, where explicit weight functions are constructed which cannot be dominated by any function $\alpha \in \mathcal{A}$.

8.9 Remark. Let $(M_p)_{p \in \mathbb{N}_0}$ be a sequence in $\mathcal{M}$ which satisfies

$$(M2) \text{ there exist } A, H > 1 \text{ so that for all } p \in \mathbb{N}: \quad M_p \leq AH^p \min_{0 \leq q \leq p} M_q M_{p-q},$$

and

$$(*) \text{ there exists } k \in \mathbb{N} \text{ so that } m_j := M_j/M_{j-1} \text{ satisfies } \liminf_{j \rightarrow \infty} \frac{m_{jk}}{m_j} > 1.$$

Then $\omega_M: [0, \infty[\rightarrow [0, \infty[$,

$$\omega_M: t \mapsto \begin{cases} \sup_{p \in \mathbb{N}_0} \log(t^p / M_p) & \text{for } t > 0 \\ 0 & \text{for } t = 0 \end{cases}$$

is equivalent to a weight function and

$$\mathcal{D}_{\{M_p\}}(\Omega) = \mathcal{D}_{\{\omega_M\}}(\Omega), \quad \mathcal{D}_{(M_p)}(\Omega) = \mathcal{D}_{(\omega_M)}(\Omega)$$

hold for each open set $\Omega \subset \mathbb{R}^N$.

To see this, note that ω_M satisfies 1.1(γ) and 1.1(δ) by definition, while 1.1(β) holds by Komatsu [14], 4.1. A result of Petzsche (see Meise [18], 2.6(2)) implies that ω_M satisfies 1.1(α). Therefore, our claim follows from Proposition 3.4 and Komatsu [14], 3.6 and 3.3.

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Eingegangen am 23. April 1990